



Deep Learning

Course code:

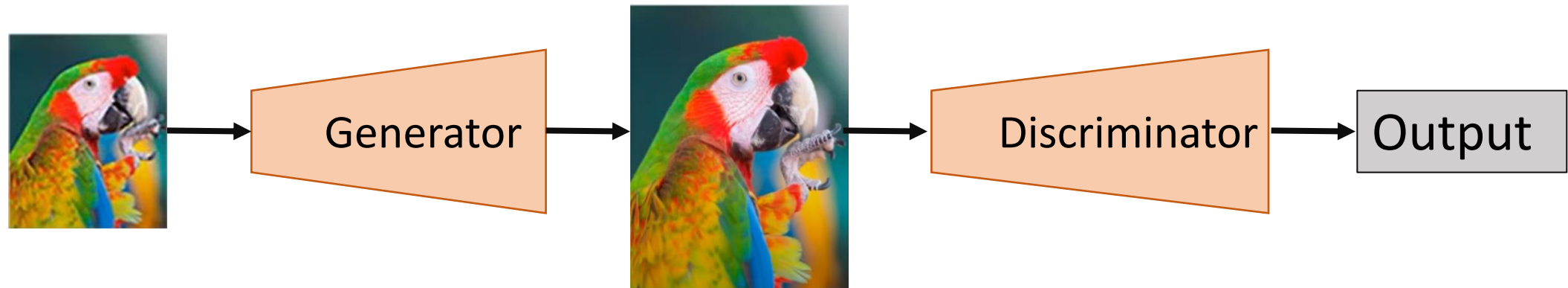
Unit 5: Autoencoders and
Generative Networks

Lecture 1: Image Data Generation
Using GAN Architectures

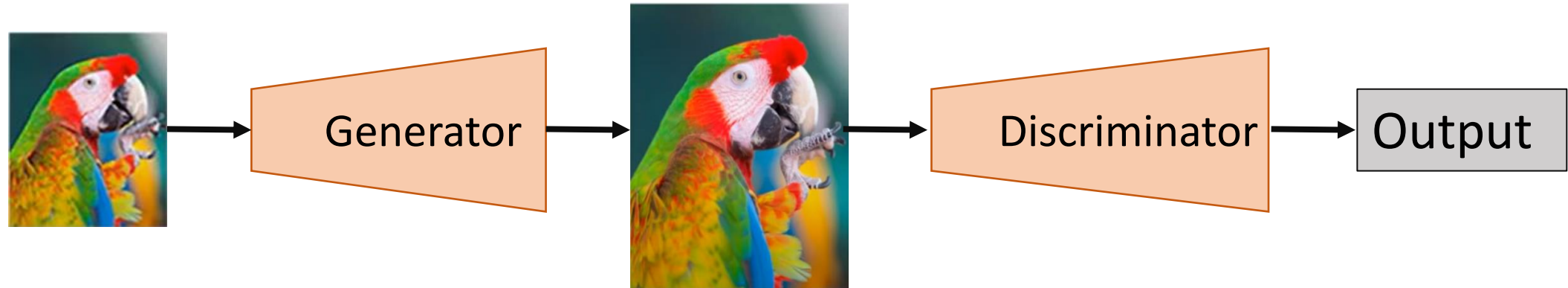




Super Resolution GAN



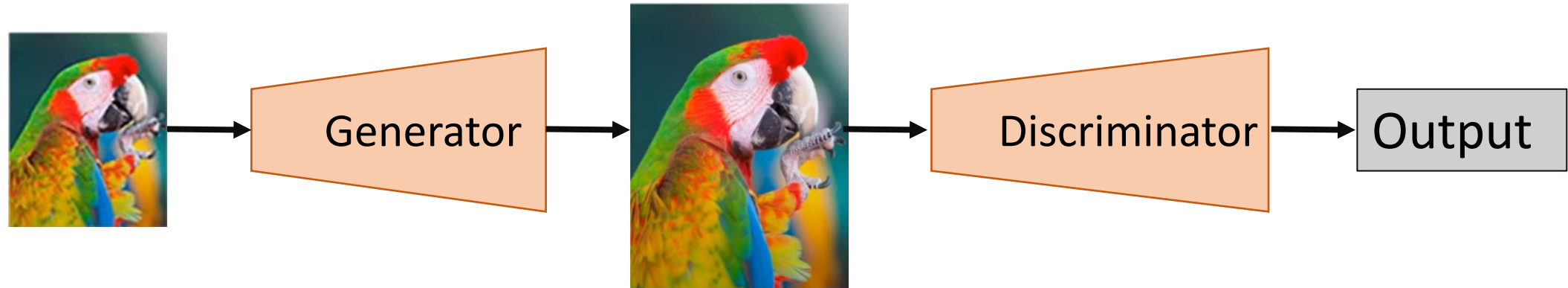
Super Resolution GAN



- Interpolation techniques for upsampling : bilinear, nearest neighbour, bicubic.

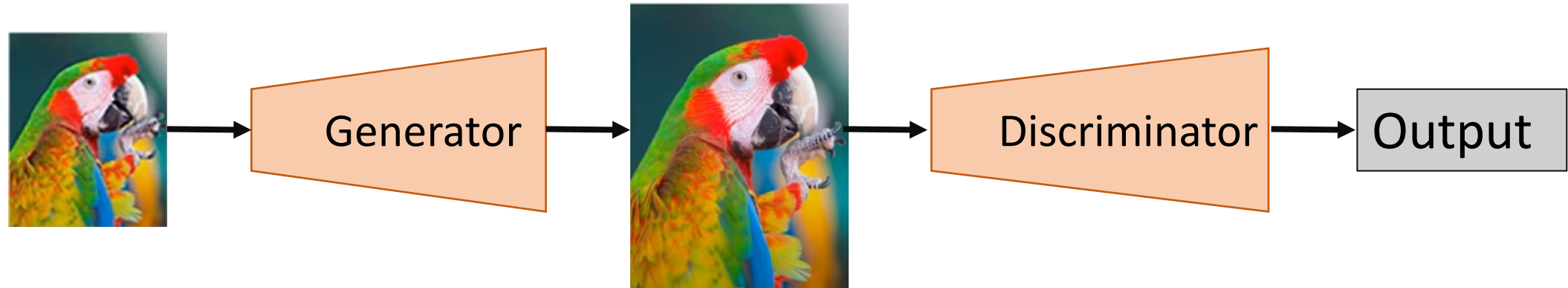


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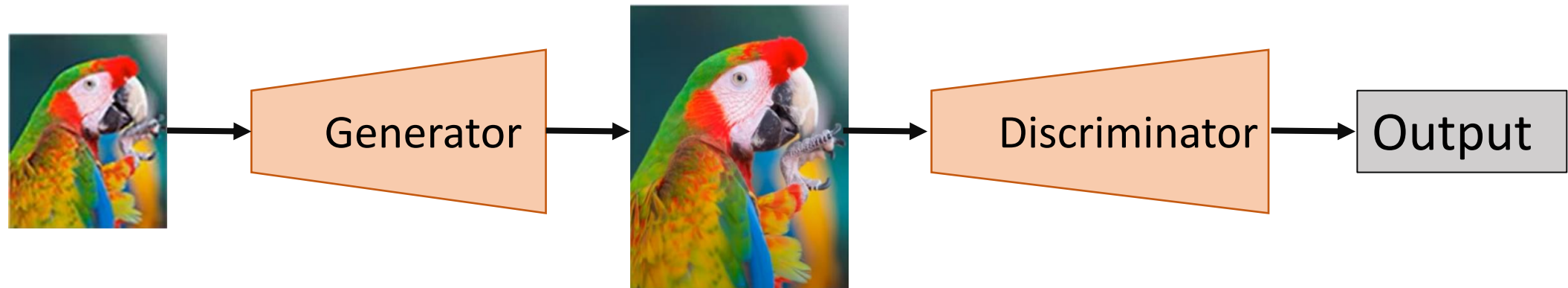
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- Super resolution ResNets: Deeper CNN for better performance.

Super Resolution GAN



- Interpolation techniques for upsampling : bilinear, nearest neighbour, bicubic.
- Super resolution using CNN: Good alternative to using traditional filters.
- Super resolution ResNets: Deeper CNN for better performance.
- SRGANs: Generative models for more realistic super resolution.



Super Resolution GAN

- **Generator:** Low resolution to super resolution image.
- **Discriminator:** Tell super resolution images from original high resolution images.



Perceptual Loss

- **Generator:** Low resolution to super resolution image.
- **Discriminator:** Tell super resolution images from original high resolution images.



Perceptual Loss:

- The perceptual loss is a combination of both adversarial loss and the content loss.

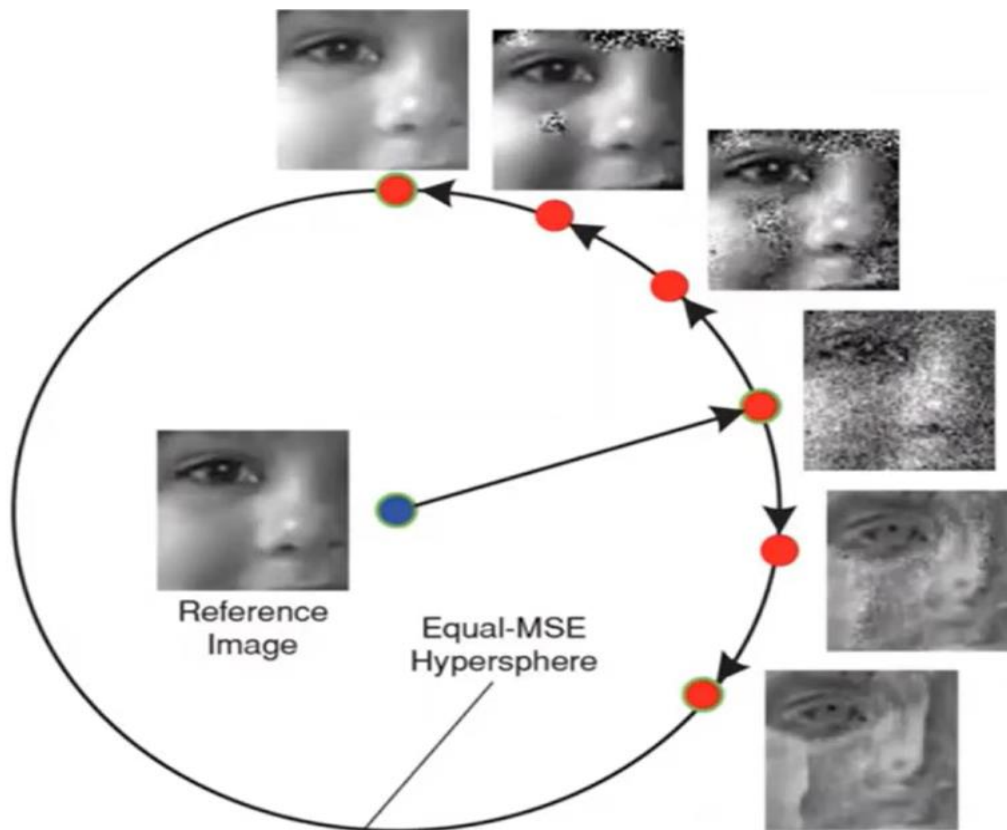
↓
Discriminator loss

↓
Content loss

Ref. <https://www.youtube.com/watch?v=KsNLxBvJBKo>

Content Loss

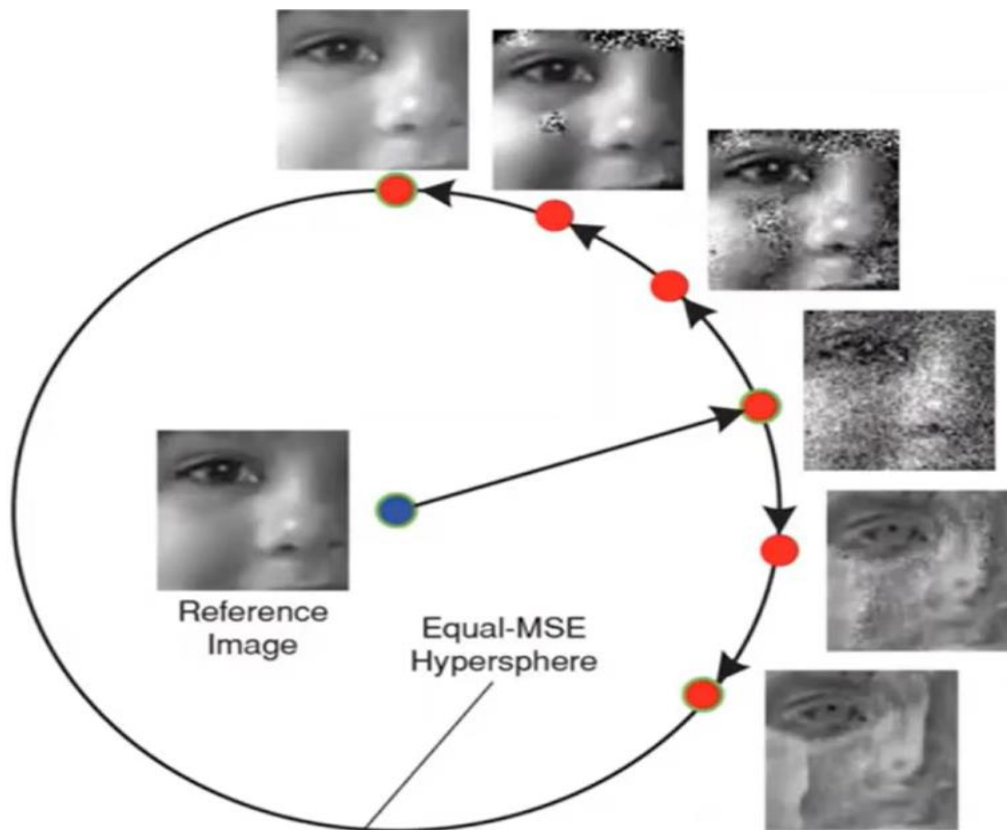
- **Content Loss:** Evaluates the image quality based on its perceptual quality.



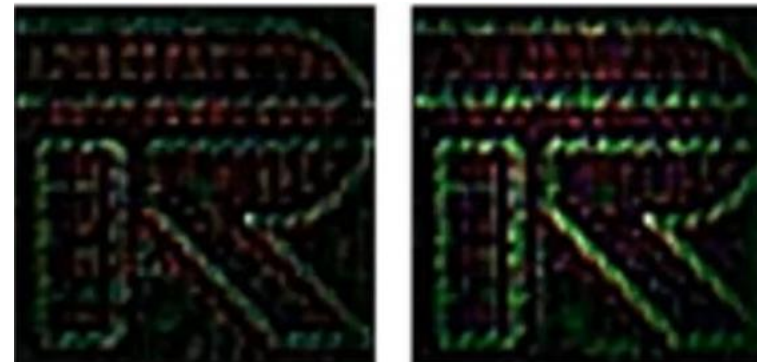
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Content Loss

- **Content Loss:** Evaluates the image quality based on its perceptual quality.



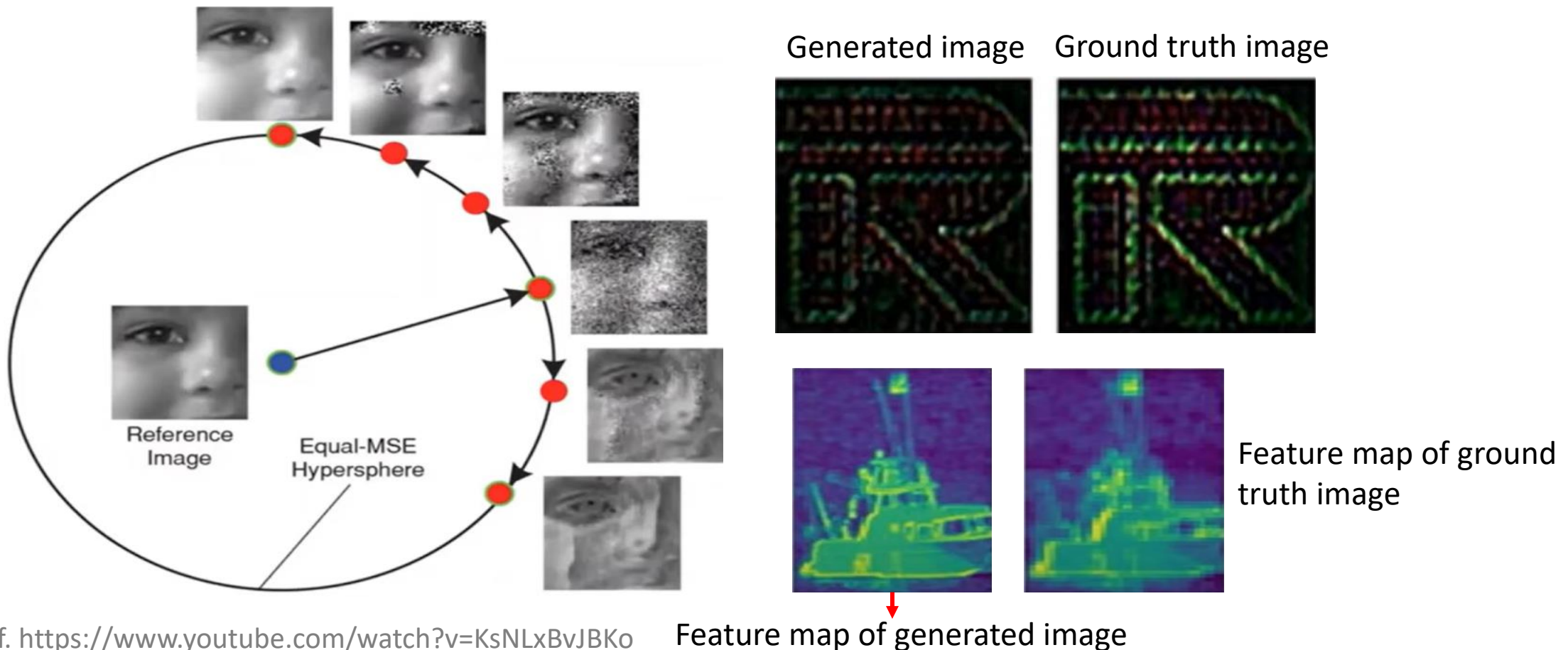
Generated image Ground truth image



Ref. <https://www.youtube.com/watch?v=KsNLxBvJBKo>

Content Loss

- **Content Loss:** Evaluates the image quality based on its perceptual quality.

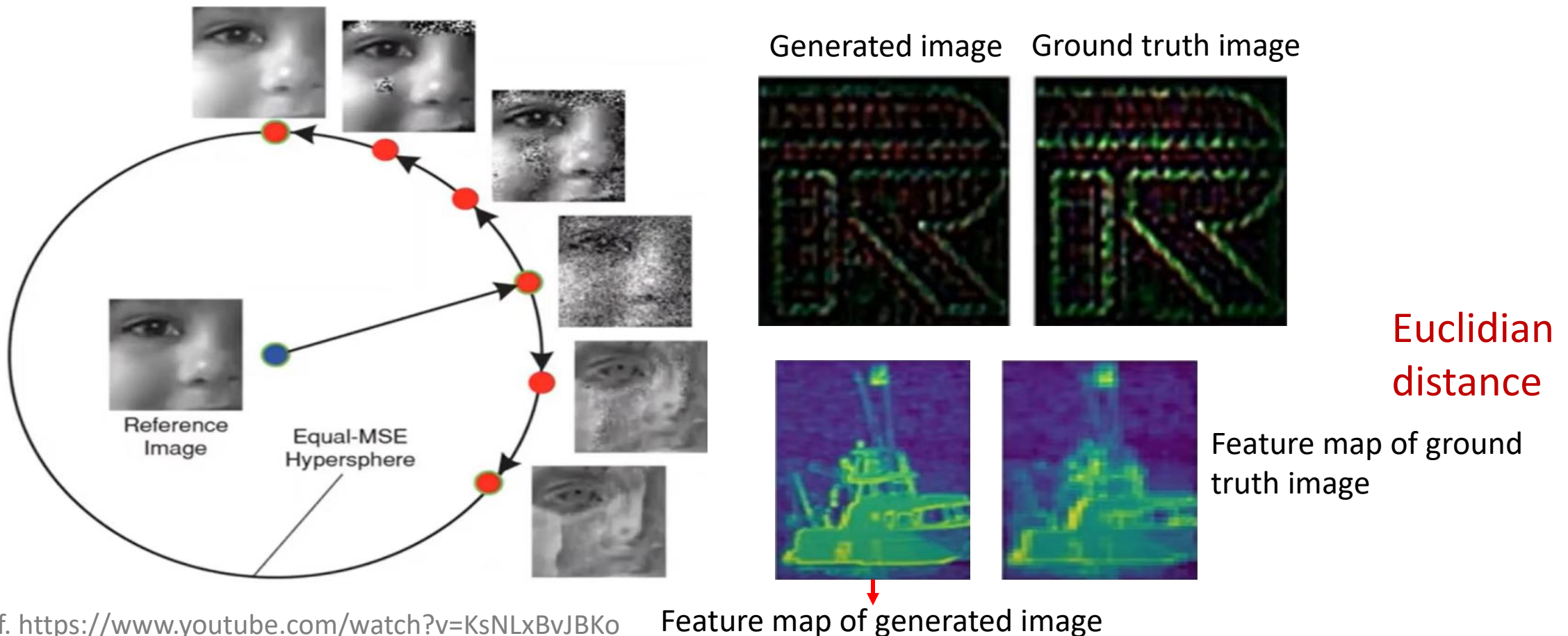


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Feature map of generated image

Content Loss

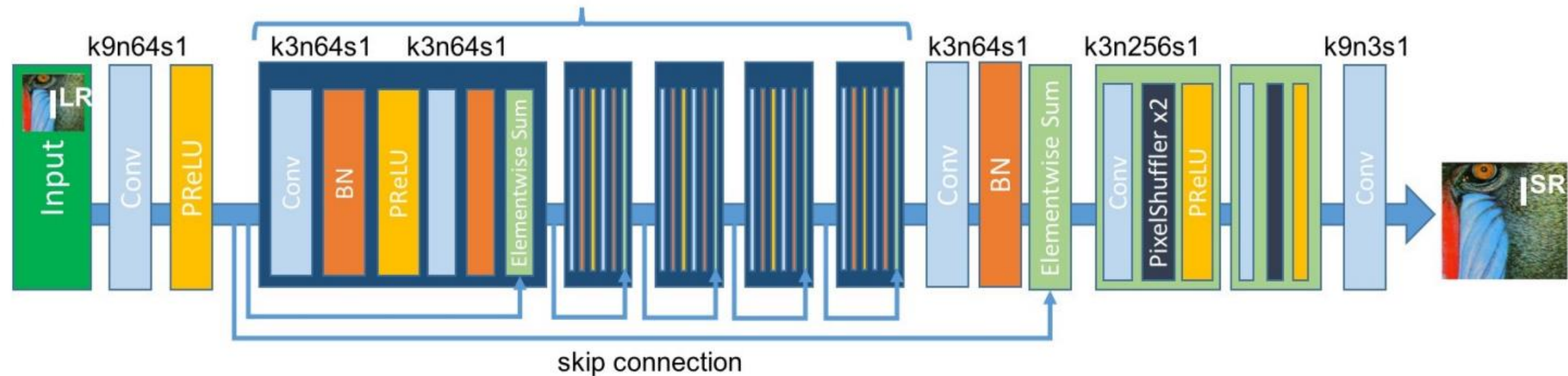
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SRGAN Generator Network

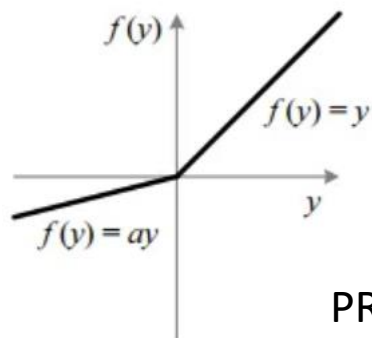
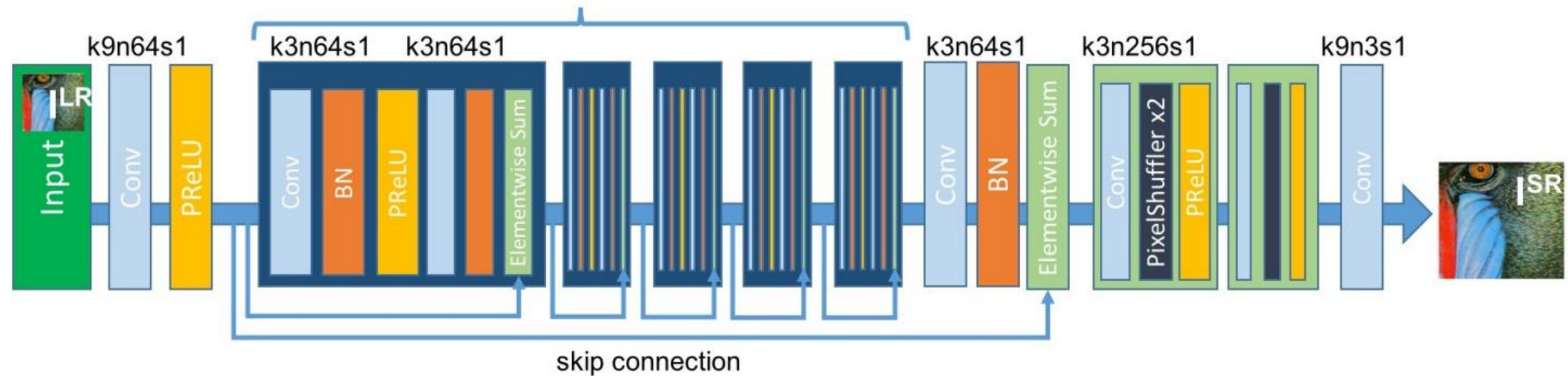
Generator Network



Ref. <https://doi.org/10.48550/arXiv.1609.04802>

SRGAN Generator Network

Generator Network

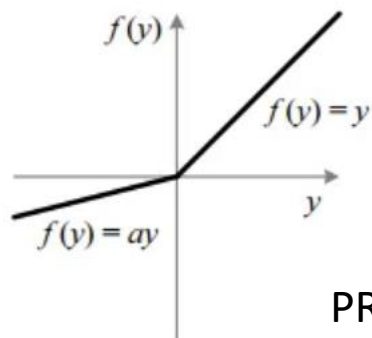
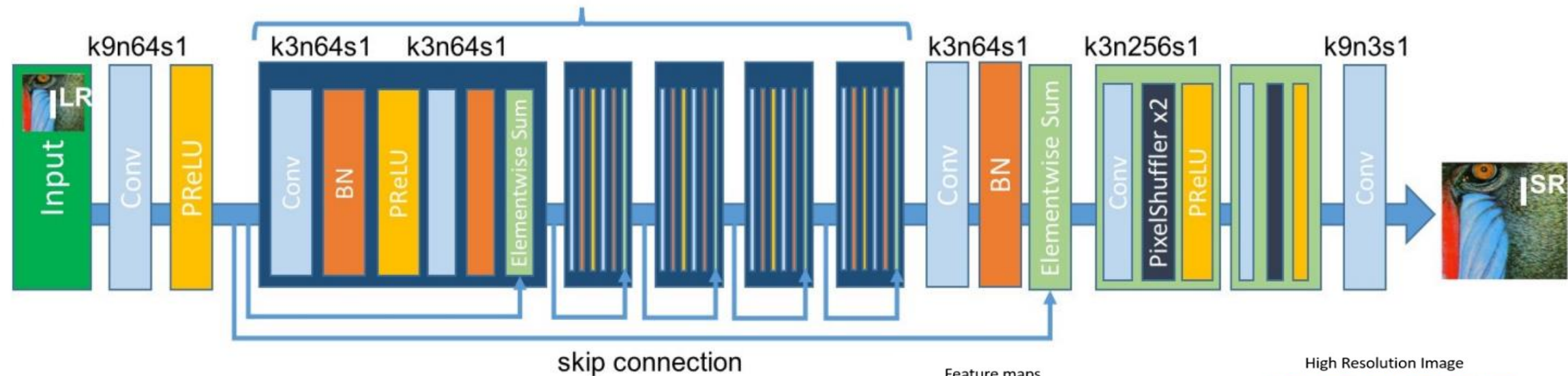


PReLU

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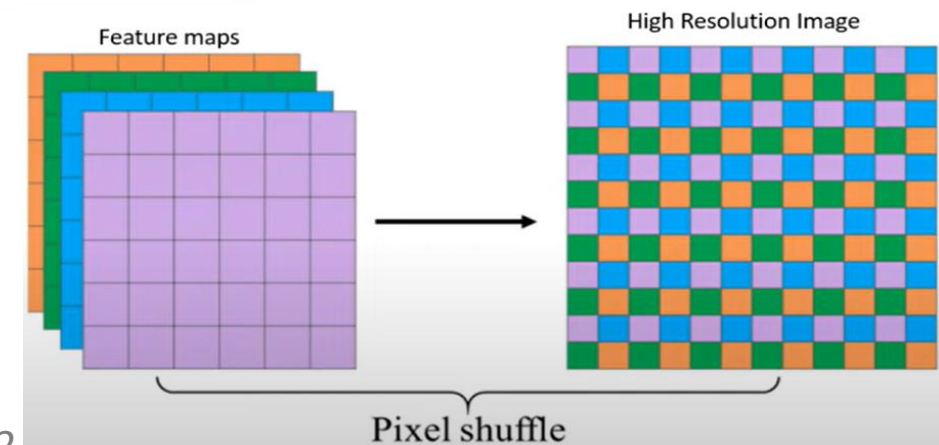
SRGAN Generator Network

Generator Network



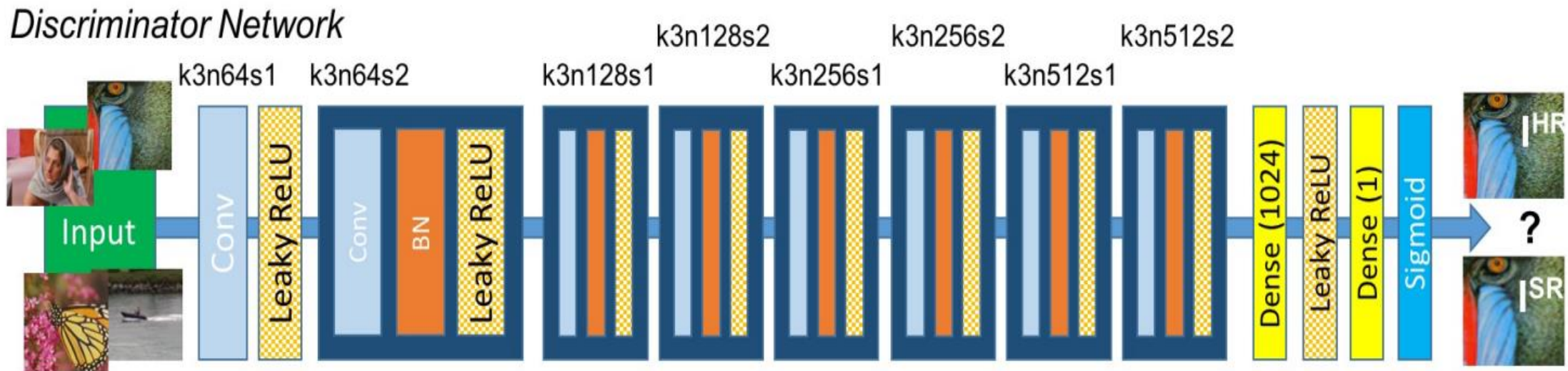
PReLU

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SRGAN Discriminator Network



Ref. <https://doi.org/10.48550/arXiv.1609.04802>



Advantages of Super Resolution GAN

- **High-quality images:** SRGAN can generate high-quality images with significantly higher resolution and finer details than the low-resolution input images.





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- **Realistic textures:** SRGAN can produce images with realistic textures, which makes the generated images more natural-looking.





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- **High-quality images:** SRGAN can generate high-quality images with significantly higher resolution and finer details than the low-resolution input images.
- **Realistic textures:** SRGAN can produce images with realistic textures, which makes the generated images more natural-looking.
- **Adaptable to different tasks:** SRGAN can be adapted to different tasks such as image restoration, denoising, and deblurring.



Disadvantages of Super Resolution GAN

- **Complexity:** SRGAN is a complex deep learning model with a large number of parameters.





Disadvantages of Super Resolution GAN

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Disadvantages of Super Resolution GAN

- **Complexity:** SRGAN is a complex deep learning model with a large number of parameters.
- **Limited training data:** SRGAN requires a large amount of high-quality training data to learn to generate high-quality images.
- **Mode collapse:** SRGAN may suffer from mode collapse, which is a common problem in GANs where the generator produces a limited set of outputs, resulting in a lack of diversity in the generated images.



References

1. Goodfellow, I., Pouget-Abadie, J., Mirza, M., Xu, B., Warde-Farley, D., Ozair, S., Courville, A. and Bengio, Y. Generative adversarial nets, NIPS (2014).
2. Goodfellow, Ian NIPS 2016 Tutorial: Generative Adversarial Networks, NIPS (2016).
3. Radford, A., Metz, L. and Chintala, S., Unsupervised representation learning with deep convolutional generative adversarial networks. arXiv preprint arXiv:1511.06434. (2015).
4. Mirza, Mehdi, and Simon Osindero. "Conditional generative adversarial nets." arXiv preprint arXiv:1411.1784 (2014).
5. Zhu, Jun-Yan, et al. "Unpaired image-to-image translation using cycle-consistent adversarial networks." Proceedings of the IEEE international conference on computer vision. 2017.
6. Ledig, Christian, et al. "Photo-realistic single image super-resolution using a generative adversarial network." Proceedings of the IEEE conference on computer vision and pattern recognition. 2017.





Deep Neural Networks





Why do we need *Sequential Modeling*





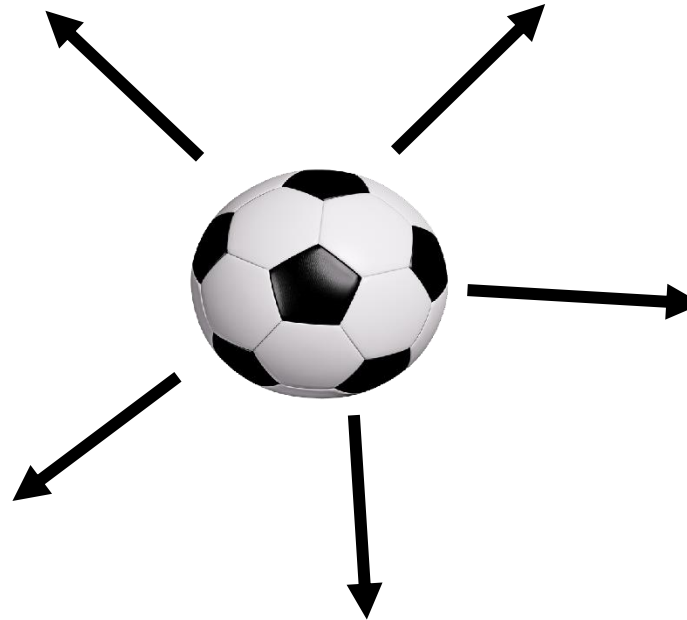
Why do we need *Sequential Modeling*

Given a Football Image





Why do we need Sequential Modeling



Given a Football Image

Can you predict where
it will go next?



Why do we need Sequential Modeling



Now can you predict?



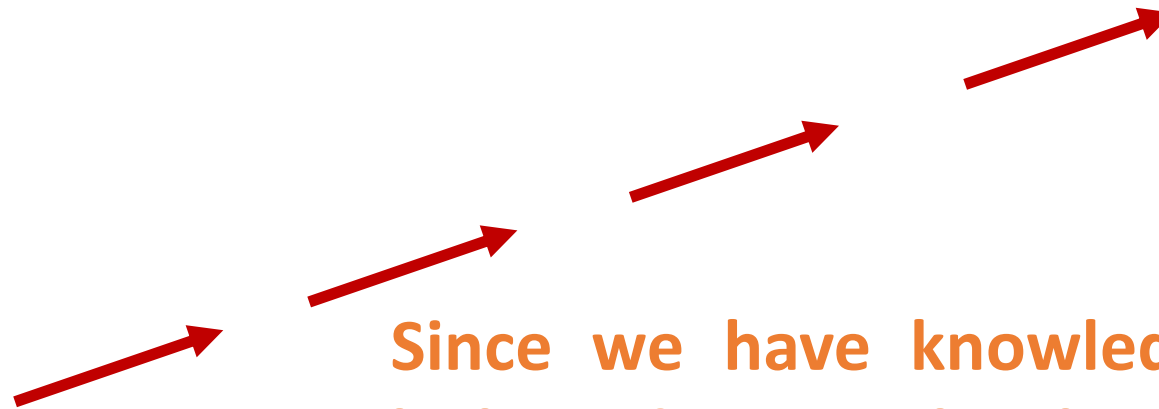
Why do we need Sequential Modeling



Since we have knowledge about the kicking direction by the player, we can predict its next direction.



Why do we need Sequential Modeling



Since we have knowledge about the kicking direction by the player, we can predict its next direction.



Need for Sequential Modeling

Sequence data

Speech recognition

Machine translation

Music generation

Name entity
recognition

Input data



You are my best friend

∅

GH Hardy said, his
contribution was
discovery of Ramanujan.

Output

Wow, it is so nice!

Você é meu melhor amigo



GH Hardy said, his
contribution was discovery
of Ramanujan.

Need for Sequential Modeling

Sequence data

Sentiment
classification

DNA sequence
analysis

Video activity
recognition

Input data

Wow, it is so nice!

```
A5ASC3.1
B4F917.1
A9S1V2.1
B9GSN7.1
Q8H056.1
Q0D4Z3.2
```



Output



```
SIKLWPPSQTTTRLLVERMANNLST..PSIFTRK.
SIKLWPPSESTRIMLVDRMTNNLST..ESIFSRK.
VFKLWPPSQGTREAVRQKMAKLSS..ACFESQS.
SVKLWPPGQSTRMLVERMTKNFIT..PSFISRK.
SFSIWPPPTQRTDRAVVRRLVDTLGG..DTILCKR.
LSIWPPSQRTDRAVVRRLVQTLVA..PSILSQR.
```

Fighting



Can we use ANN/CNN for *Sequential Modeling*?



Can we use ANN/CNN for *Sequential Modeling*?

No 

Reasons:

- Fixed input size

Example : **image size**



32x32

Reasons:

- Fixed input size

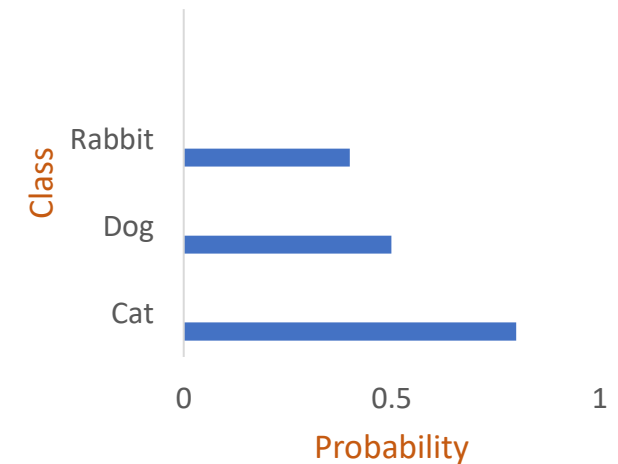
Example : image size



32x32

- Fixed output size

Example : probabilities of different classes



Reasons:

- Fixed input size

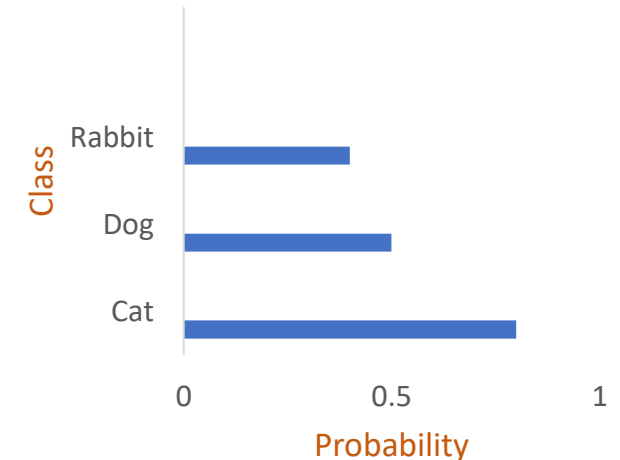
Example : **image size**



32x32

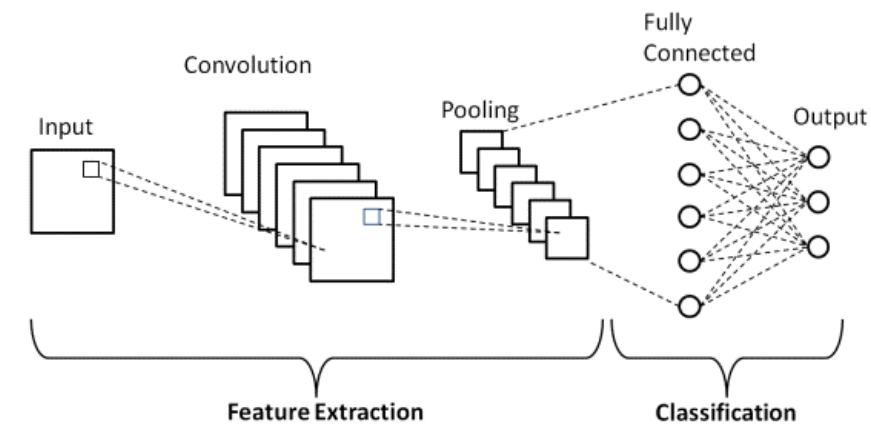
- Fixed output size

Example : **probabilities of different classes**



- Fixed computational steps

Example : **number of layers in the model**





Reasons:

- Words learned or approximated at a later position may change the approximation of a previous word.

Example :

- Blue dresses are looking good.
 - Blue dress is looking good.
- Parameter sharing is not done in conventional ANNs.





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There comes ***Recurrent Neural Network!***



Reasons for Using Recurrent Neural Network (RNN)

- Can handle inputs and outputs of **varying lengths**.





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- **Shares parameters** across different parts of the network.
- Track **long-term** dependencies.





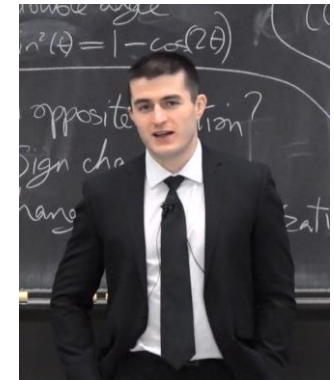
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- It involves **directed cycles** to recognize sequential characteristics of a data.
- **Shares parameters** across different parts of the network.
- Track **long-term** dependencies.
- Maintain information about **order**.



When to use RNN?

“Whenever there is a sequence of data and the temporal dynamics that connects the data is more important than the spatial content of each individual frame.”

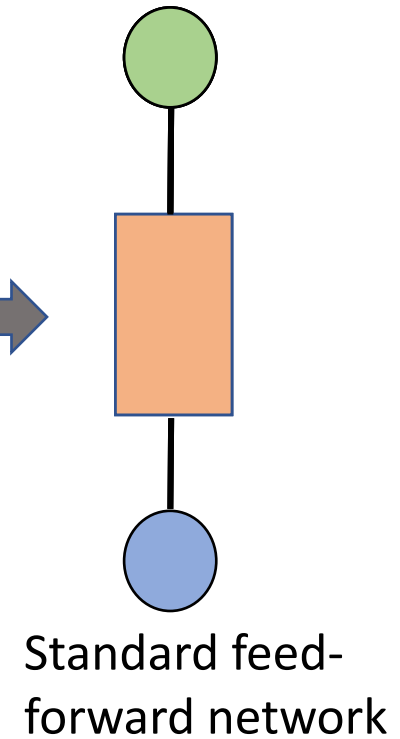
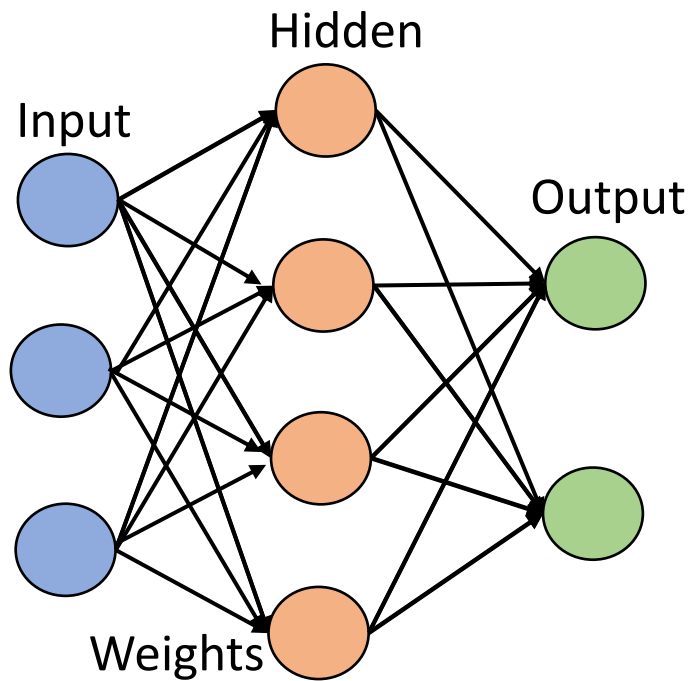


– Lex Fridman (MIT)

Image source: https://commons.wikimedia.org/wiki/File:Lex_Fridman_teaching_at_MIT_in_2018.png



Neural Network: Simplified



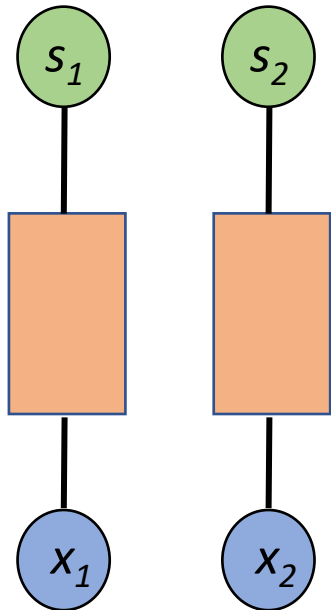


Handling Individual Time Steps



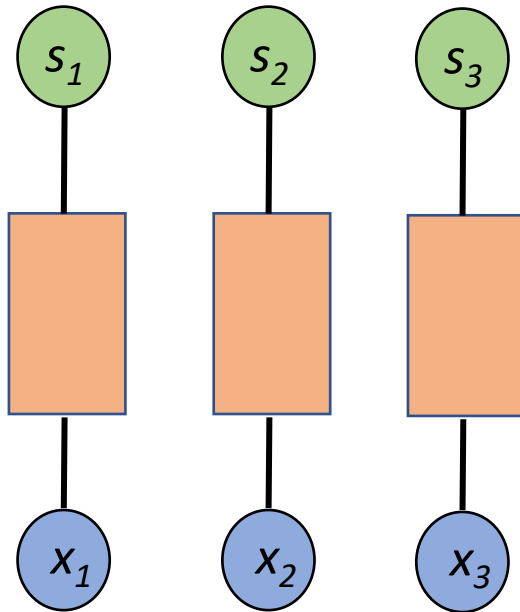


Handling Individual Time Steps

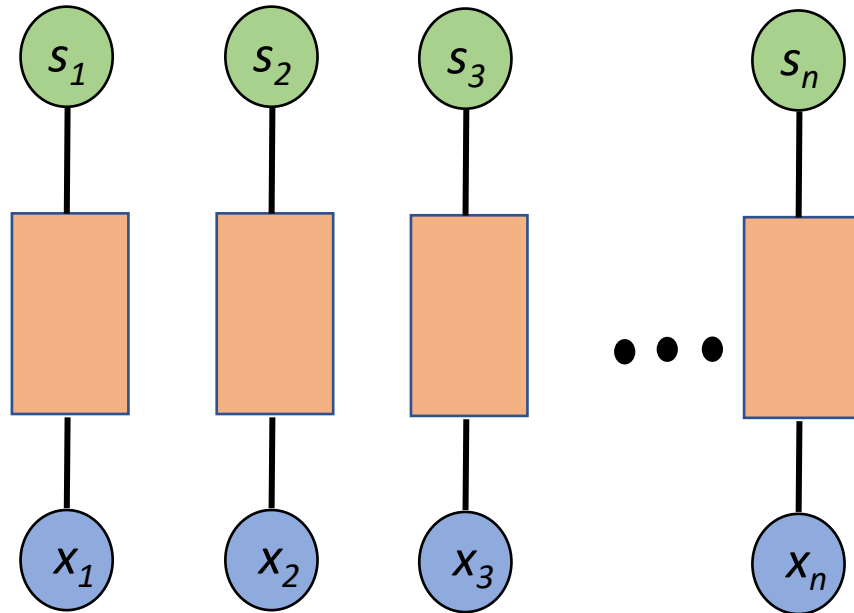




Handling Individual Time Steps



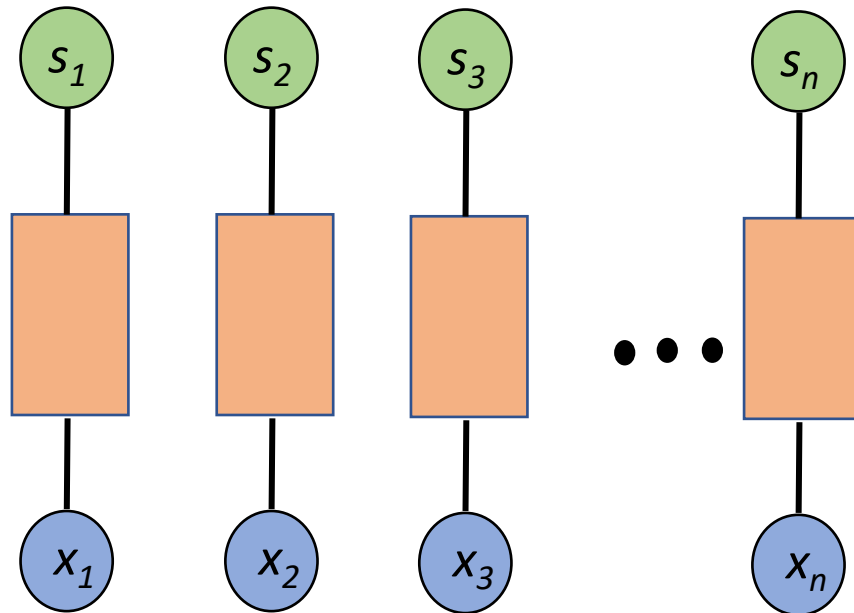
Handling Individual Time Steps



We can notice here that all these models are simply copying the same thing with different inputs at different time steps. We can make it concrete in terms of function transformation, since the predicted output is a function of input vector, hidden state, and the weight matrix



Handling Individual Time Steps



In general,

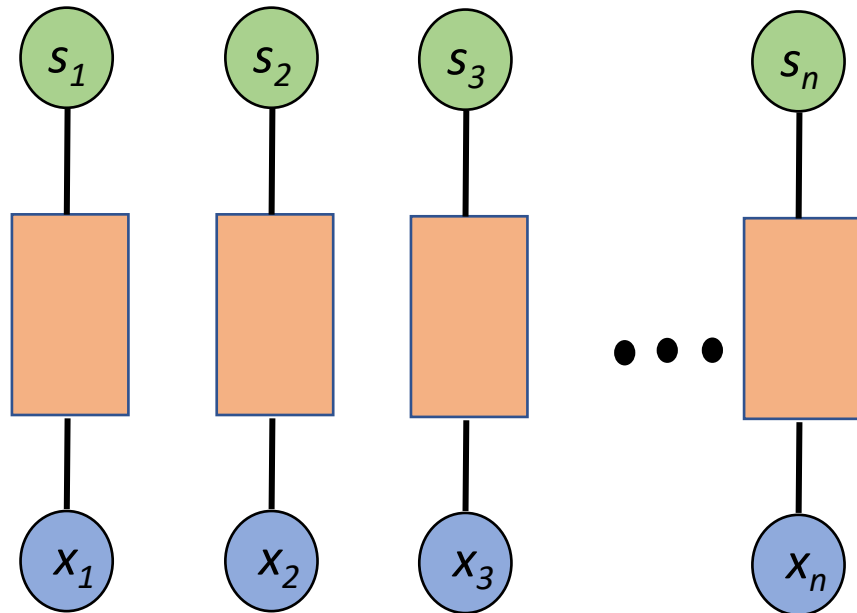
$$s_n = f(x_n)$$

$n = \text{time step}$

Note: We can make it concrete in terms of function transformation, since the predicted output is a function of input vector. The function f is learned by the weights of the network.



Handling Individual Time Steps



In general,

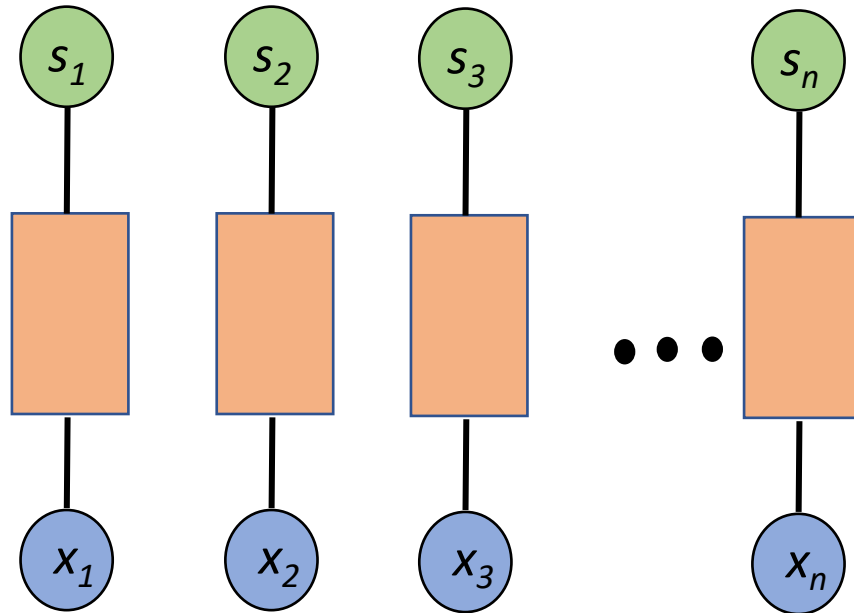
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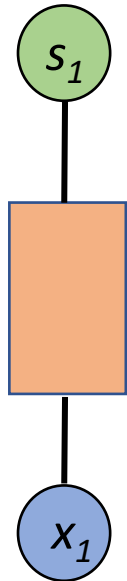
- Same function is used
- Replicate network any number of times
- Ensure parameter sharing
- Number of timestep does not matter



Handling Individual Time Steps



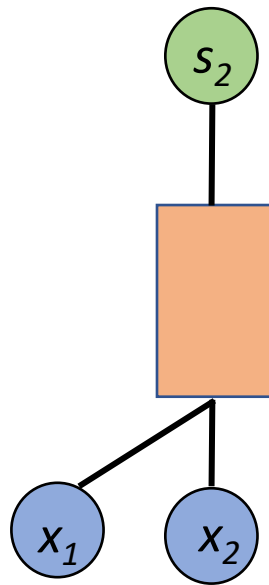
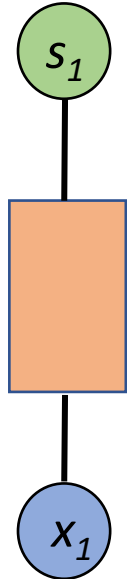
How to maintain the
interdependency between input?



A Simple Approach

Let's consider one approach





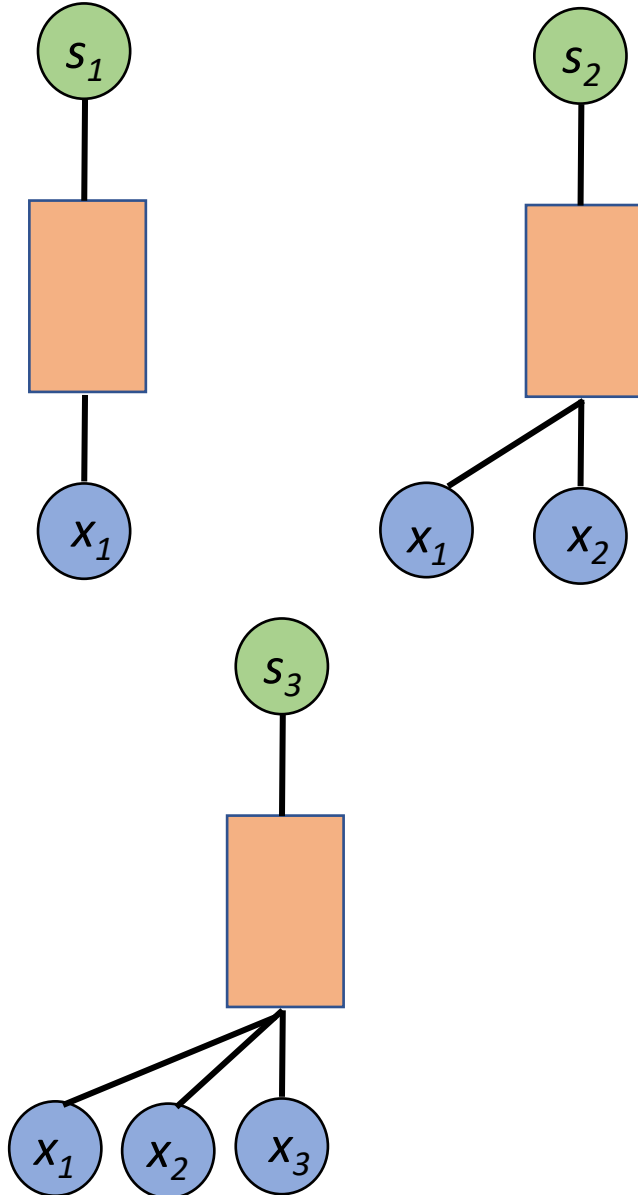
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A Simple Approach

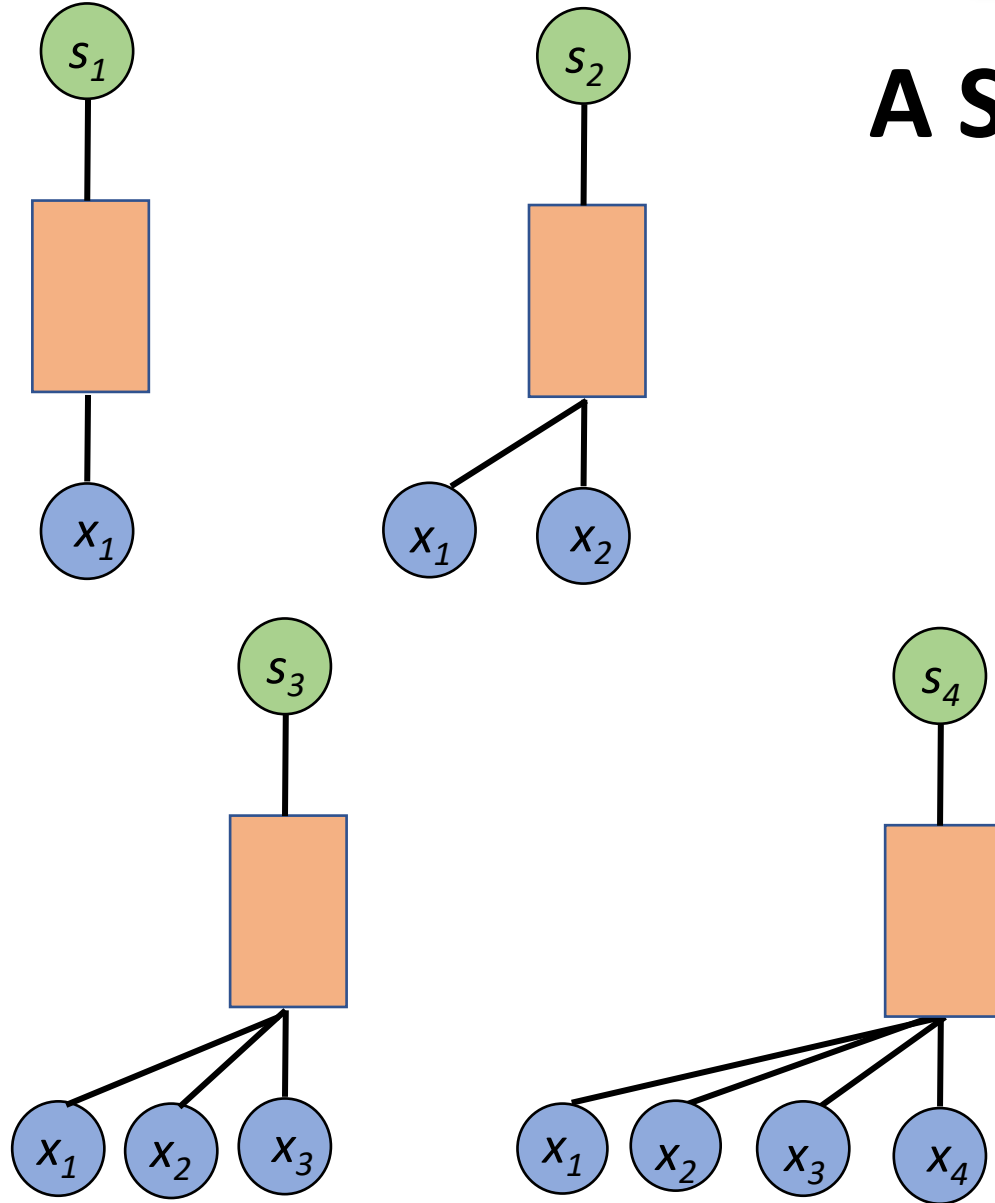
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A Simple Approach

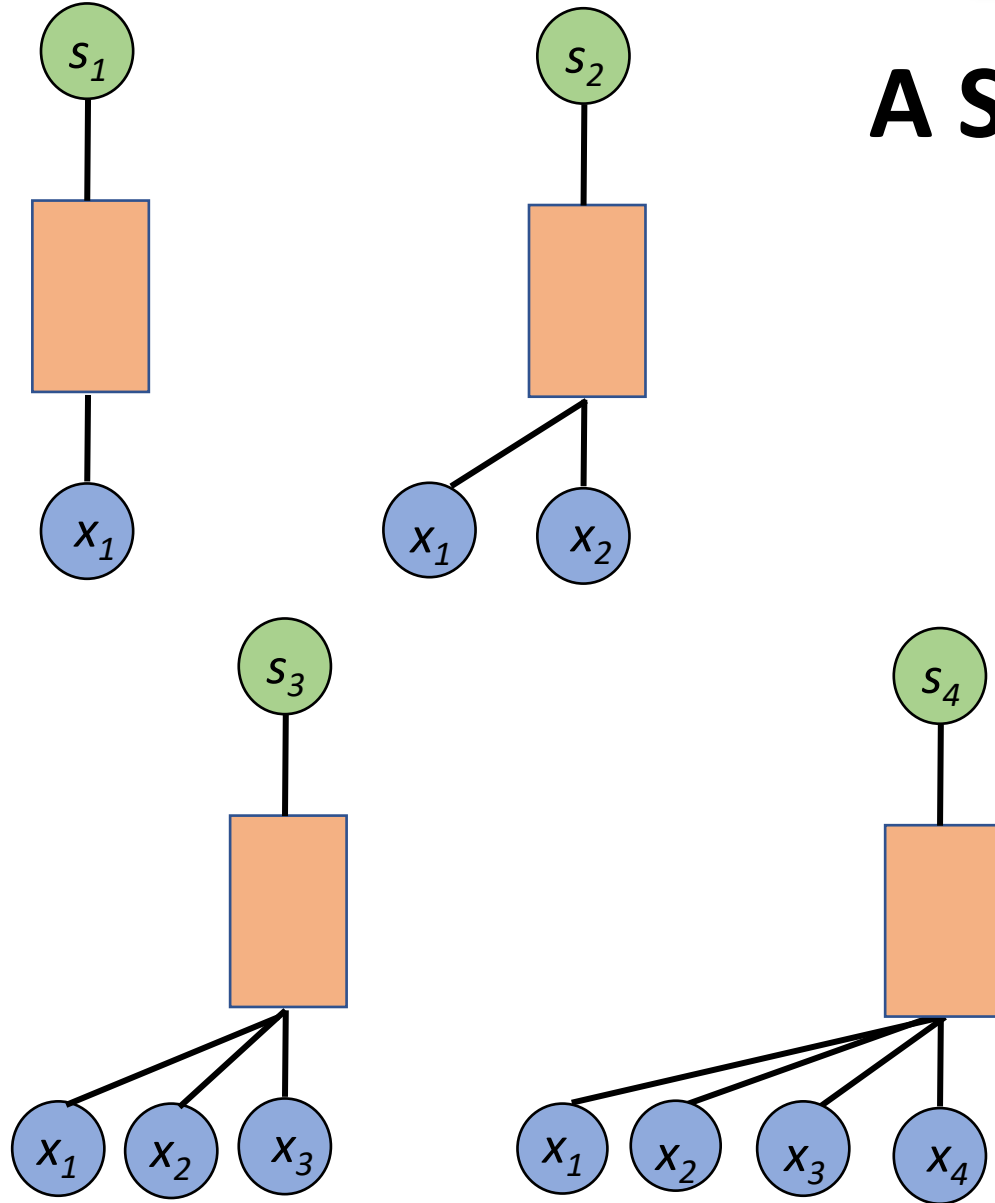
Let's consider one approach





A Simple Approach

Will this approach work?





A Simple Approach

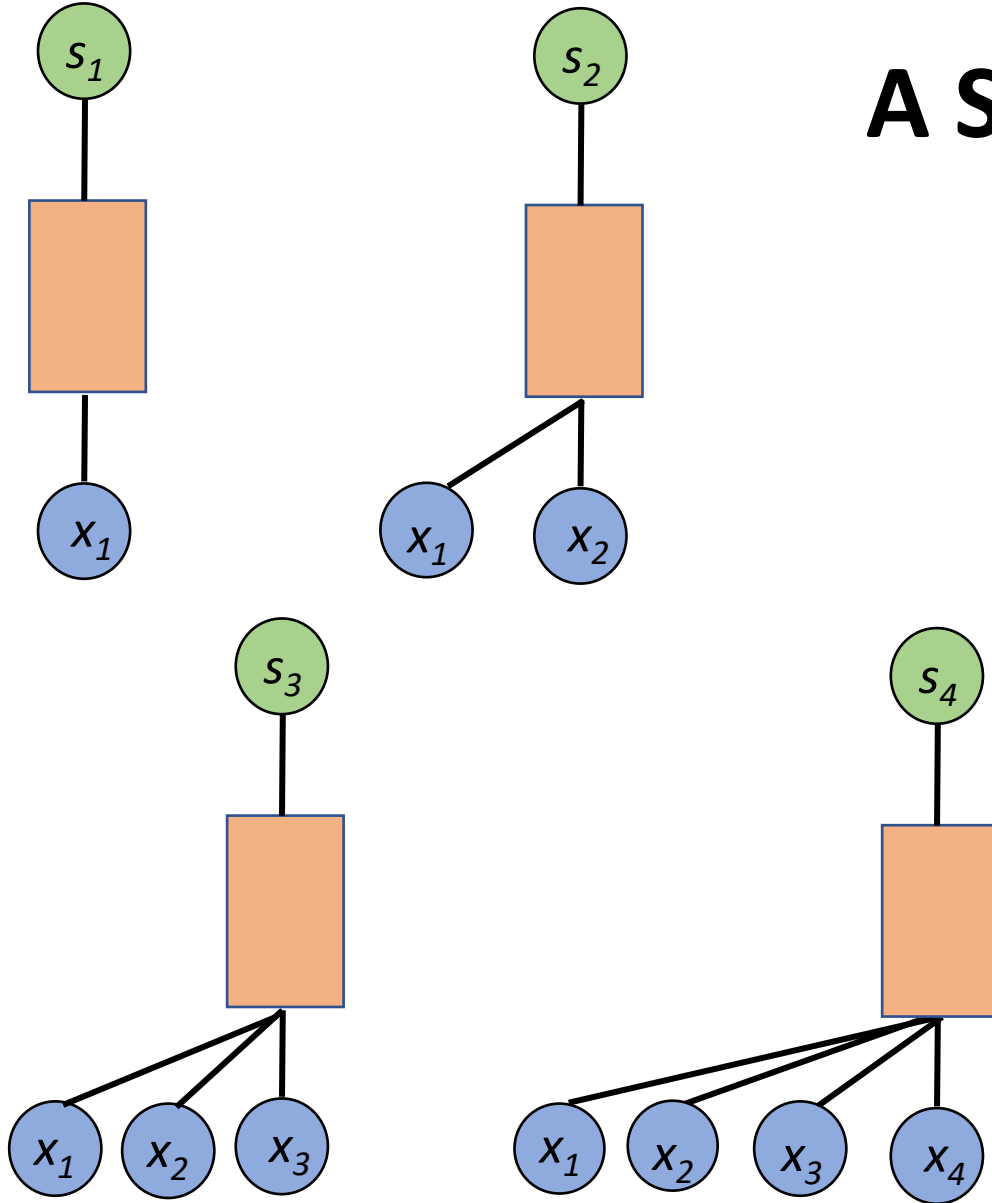
Problem

- Different function for different time-step

$$s_1 = f_1(x_1)$$

$$s_2 = f_2(x_1, x_2)$$

$$s_3 = f_3(x_1, x_2, x_3) \dots\dots$$





A Simple Approach

Problem

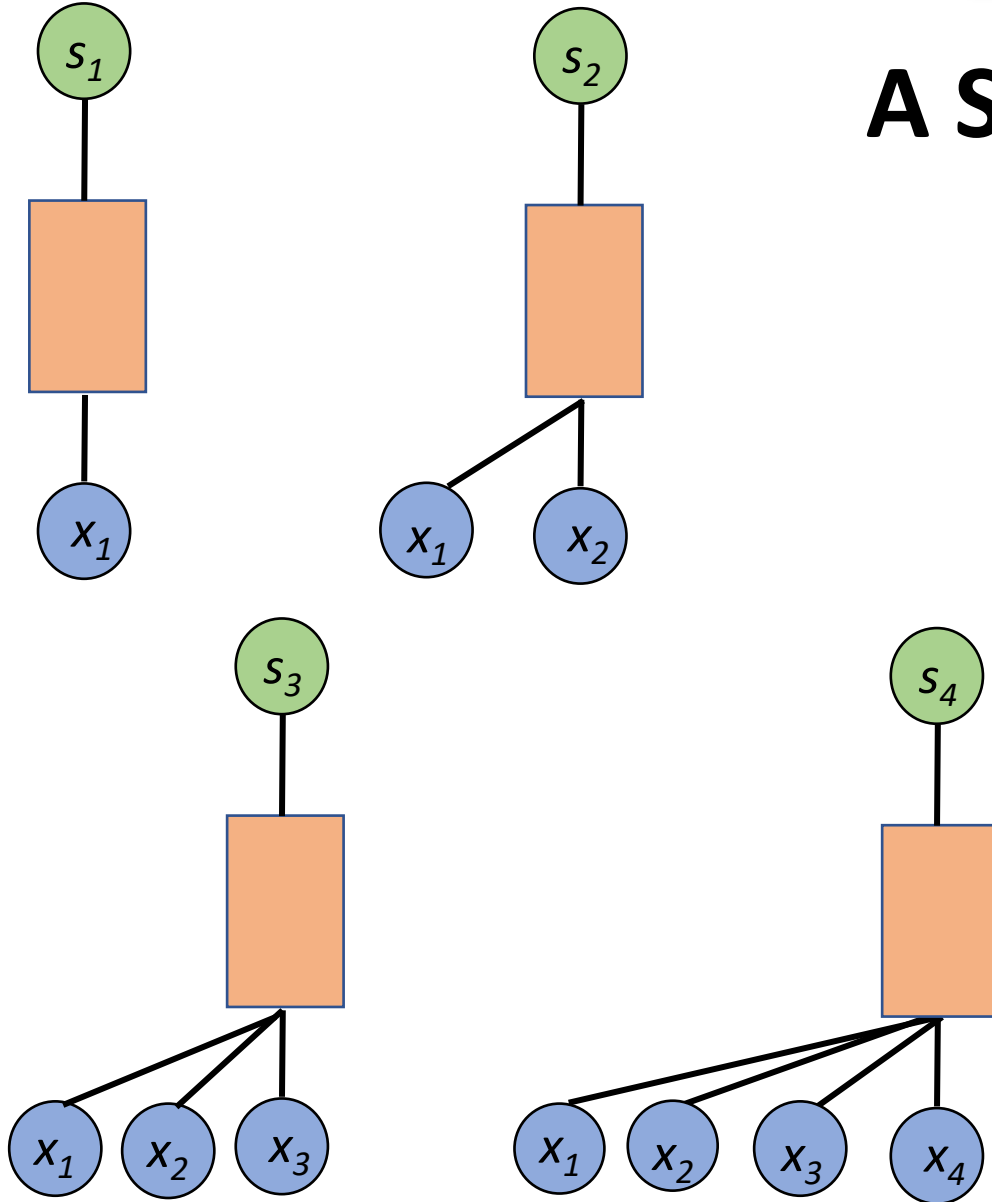
- Different function for different time-step

$$s_1 = f_1(x_1)$$

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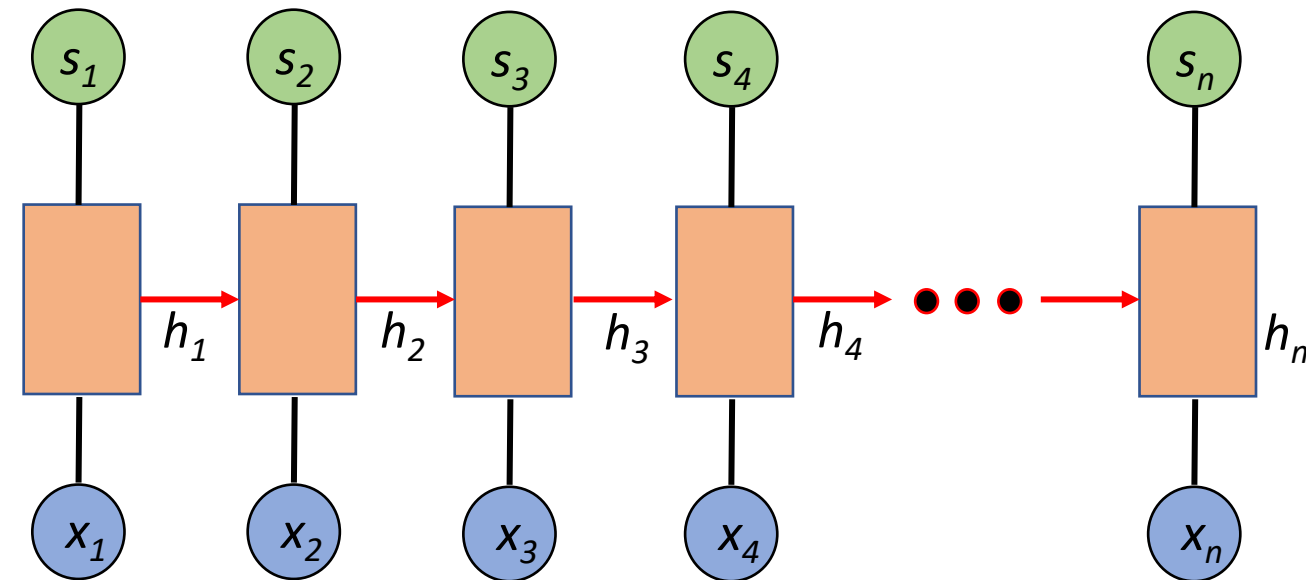
$$s_3 = f_3(x_1, x_2, x_3) \dots\dots$$

- Depends on input length





Recurrent Neural Network



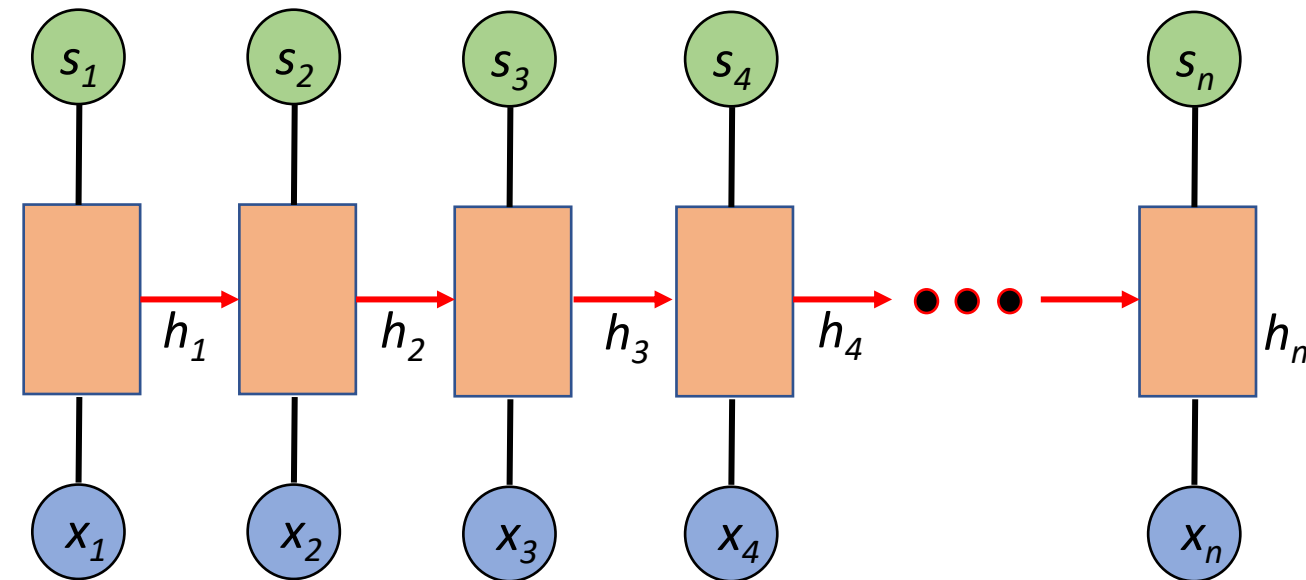
Solution

Add recurrent connection

Solution of interdependency is to add a recurrent connection in the network.
A sequence of vectors is processed by applying a recurrence formula at each time step.



Recurrent Neural Network



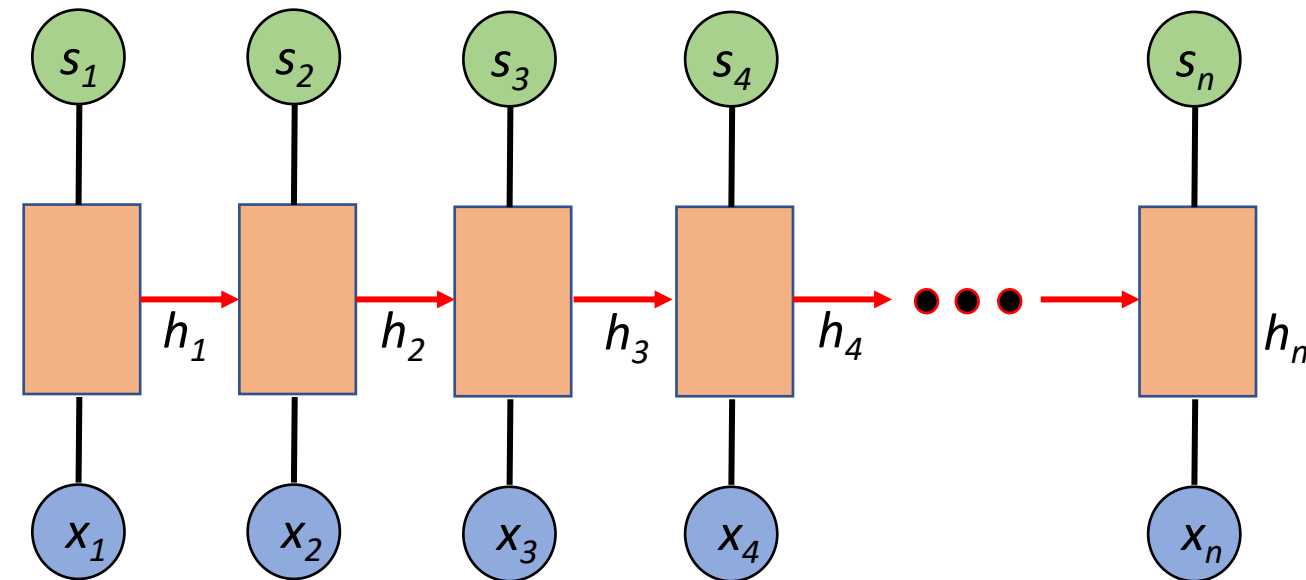
Solution

Add recurrent connection

$$s_n = f(x_n, h_{n-1})$$



Recurrent Neural Network



Solution

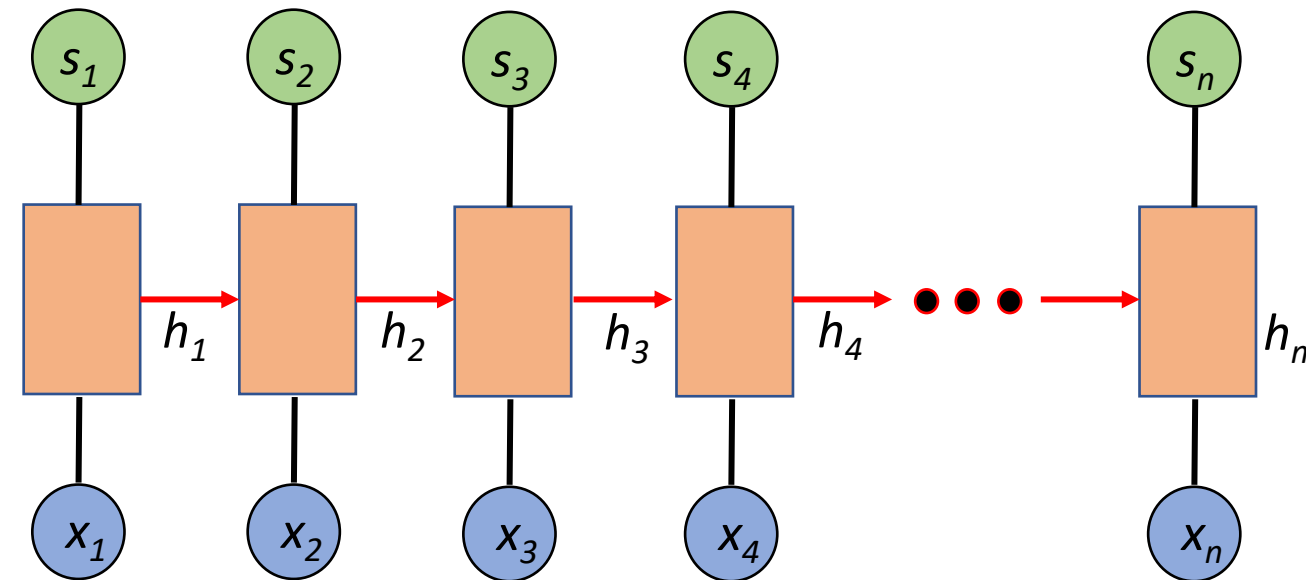
Add recurrent connection

$$s_n = f(x_n, h_{n-1})$$

input



Recurrent Neural Network



Solution

Add recurrent connection

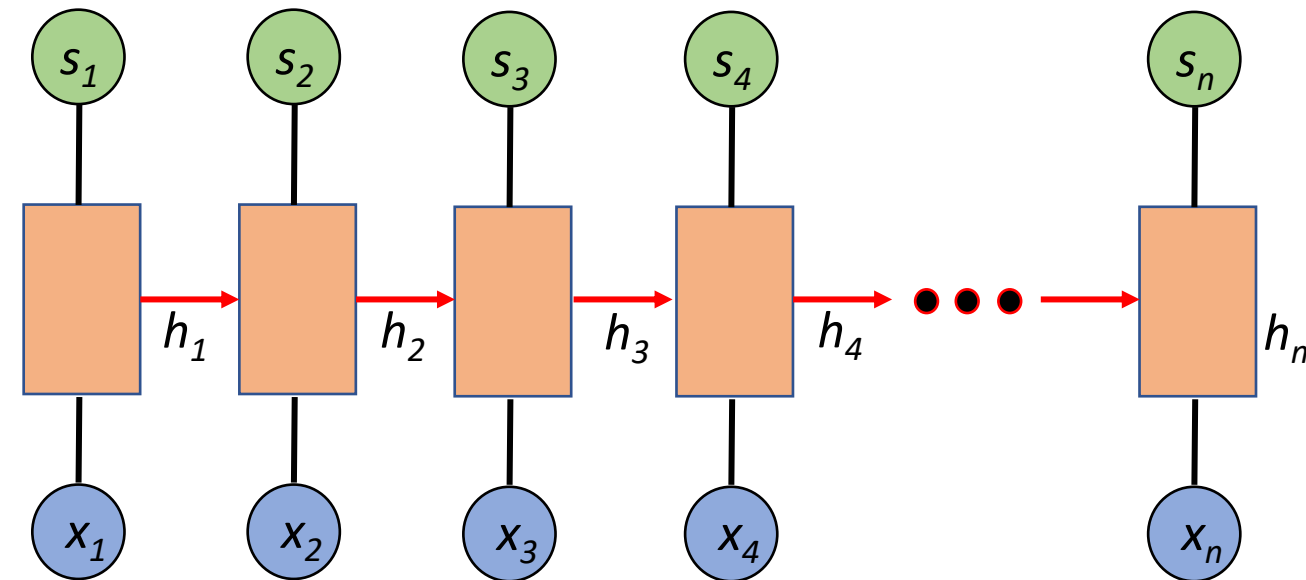
$$s_n = f(x_n, h_{n-1})$$

output

input



Recurrent Neural Network



Solution

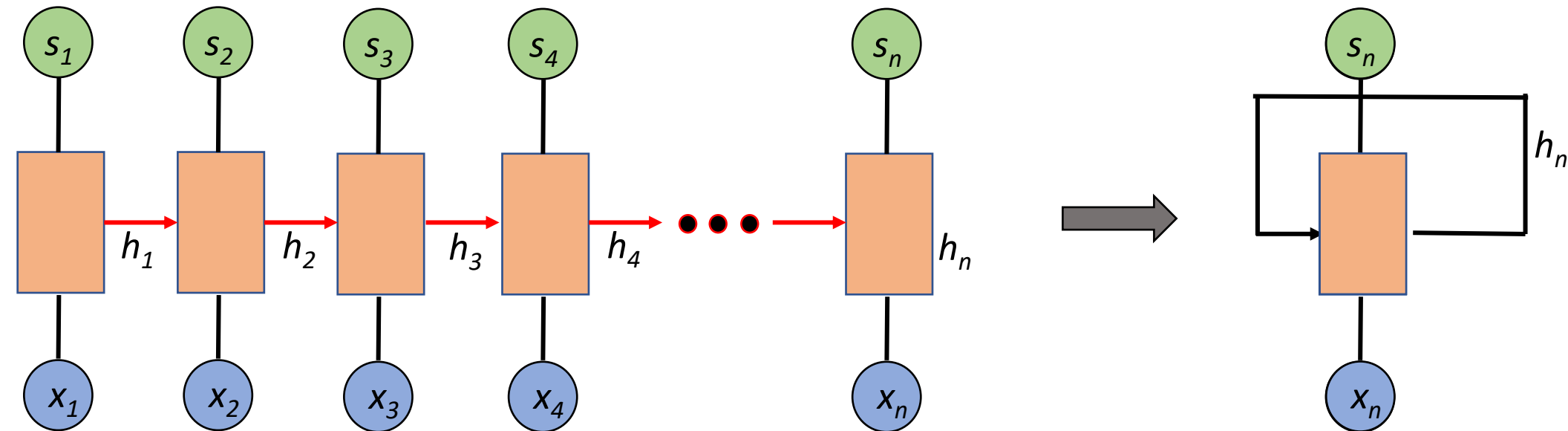
Add recurrent connection

$$s_n = f(x_n, h_{n-1})$$

output input past memory



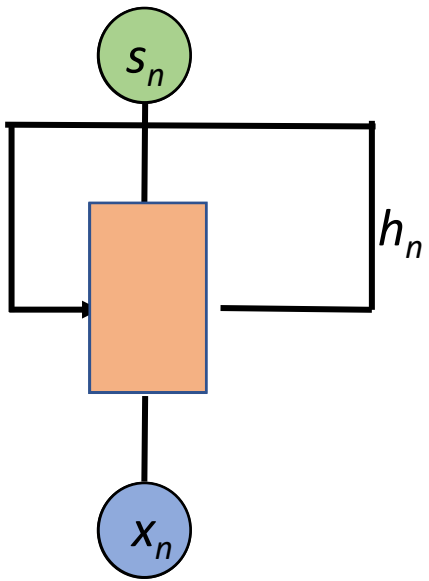
Recurrent Neural Network



Can be represented
more compactly

Recurrent Neural Network

A sequence of vectors is processed by applying a recurrence formula at each time step.

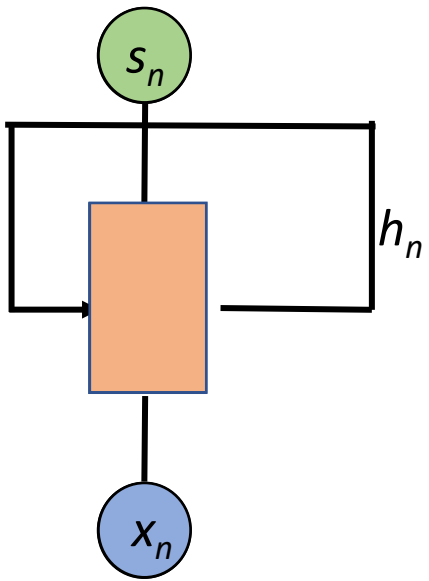




Recurrent Neural Network

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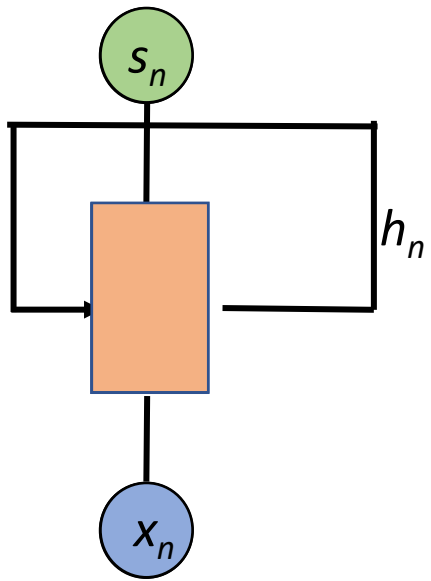
$$h_n = f_W(x_n, h_{n-1})$$



At each time step, h_n is updated as a sequence of input is processed.
The same function and same set of parameters are used in each time step.

Recurrent Neural Network

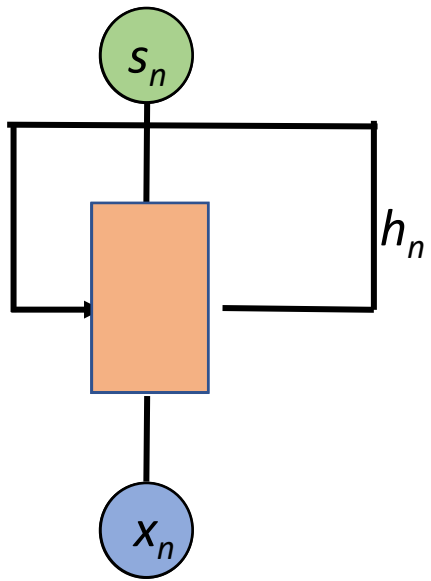
A sequence of vectors is processed by applying a recurrence formula at each time step.



Hidden state $\rightarrow h_n = f_W(x_n, h_{n-1})$

Recurrent Neural Network

A sequence of vectors is processed by applying a recurrence formula at each time step.



$$h_n = f_W(x_n, h_{n-1})$$

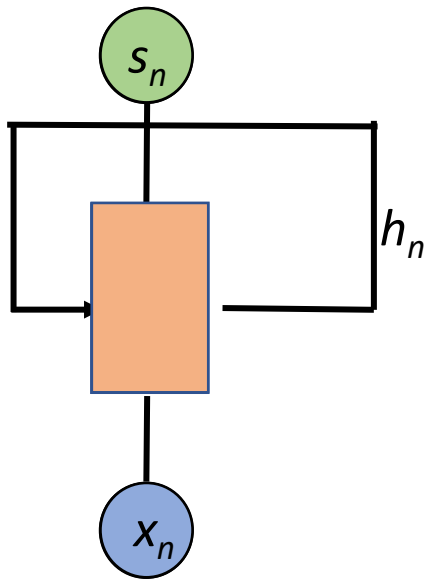
Hidden state

Function with weight W



Recurrent Neural Network

A sequence of vectors is processed by applying a recurrence formula at each time step.



$$h_n = f_W(x_n, h_{n-1})$$

Hidden state

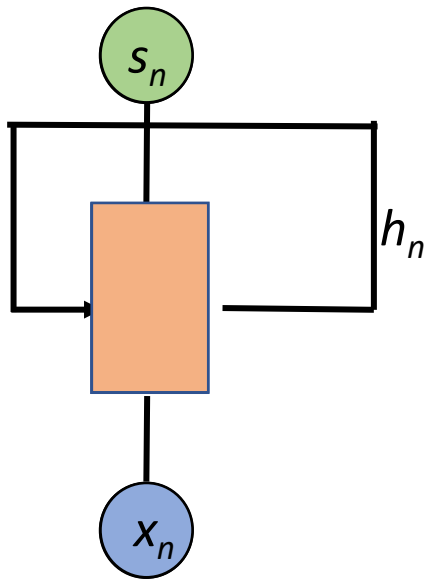
Function with weight W

Input



Recurrent Neural Network

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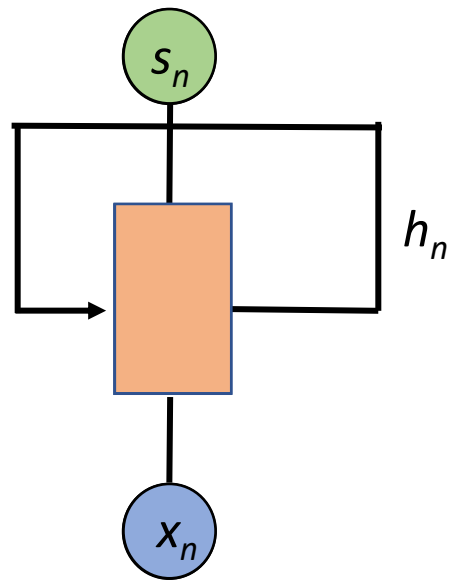
$$h_n = f_W(x_n, h_{n-1})$$

Hidden state → h_n
Function with weight W → f_W
Input → x_n
Old hidden state → h_{n-1}



Recurrent Neural Network

A sequence of vectors is processed by applying a recurrence formula at each time step.



$$h_n = f_W(x_n, h_{n-1})$$

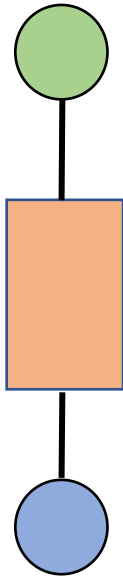
Hidden state h_n is calculated using the function f_W (Function with weight W) applied to the current input x_n and the old hidden state h_{n-1} .

- Same function is used
- Ensure parameter sharing
- Handles the temporal dependency between sequence



Recurrent Neural Network Architectures

one to one



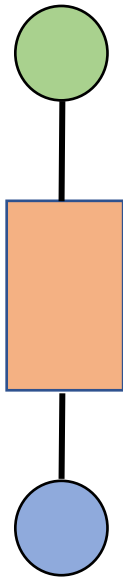
Vanilla NN





Recurrent Neural Network Architectures

one to one



Vanilla NN

one to many

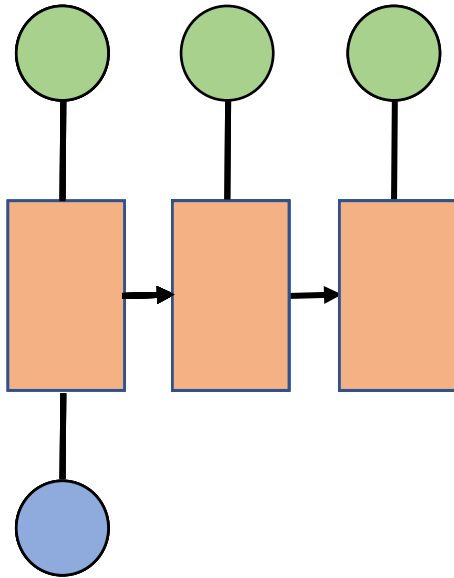


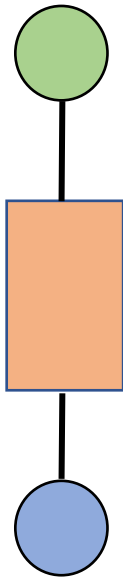
Image
Captioning





Recurrent Neural Network Architectures

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Vanilla NN

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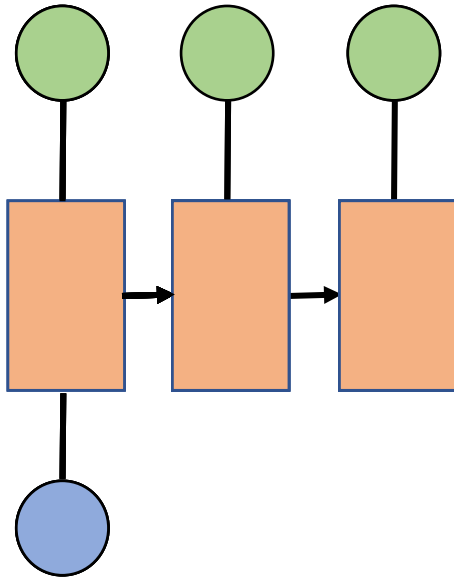
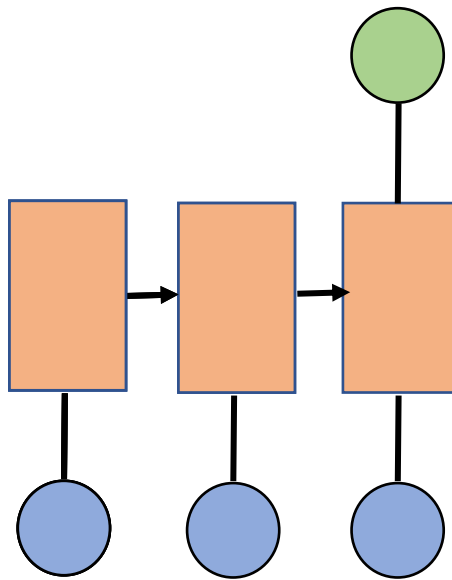


Image
Captioning

many to one

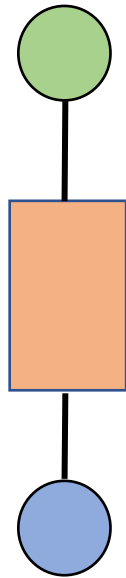


Sentiment
Classification



Recurrent Neural Network Architectures

one to one



Vanilla NN

one to many

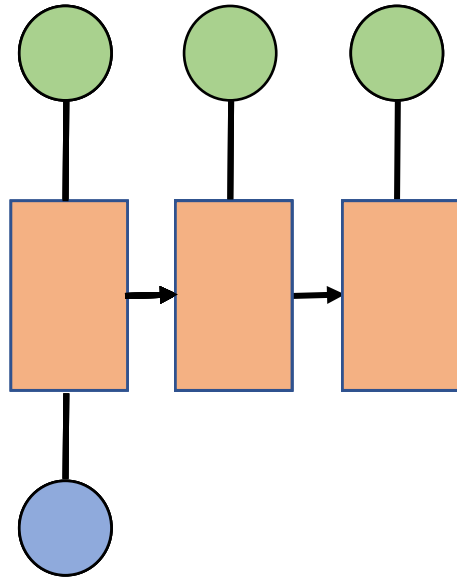
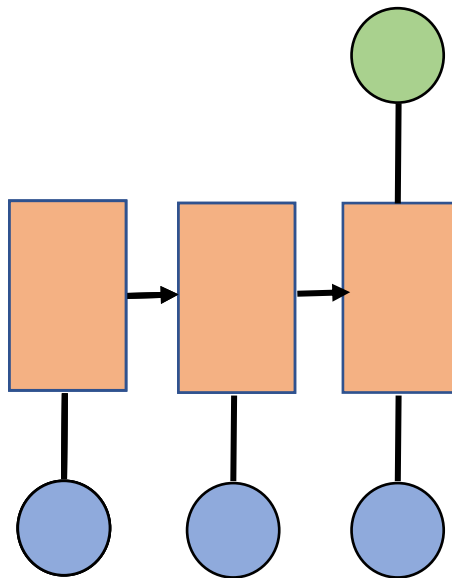


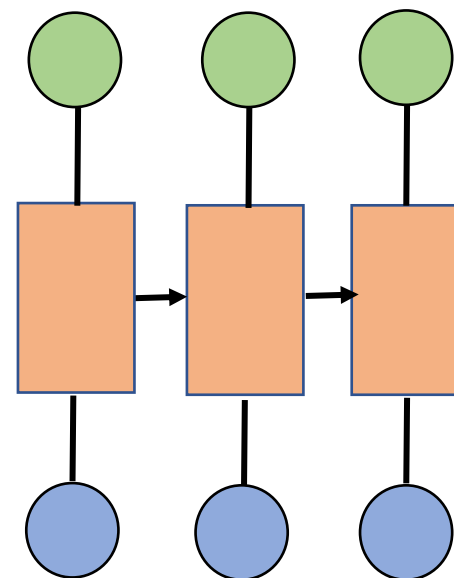
Image
Captioning

many to one



Sentiment
Classification

many to many

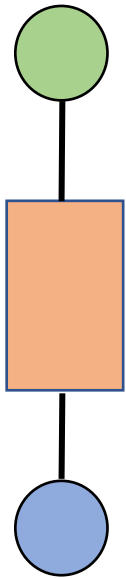


Name Entity
recognition



Recurrent Neural Network Architectures

one to one



Vanilla NN

one to many

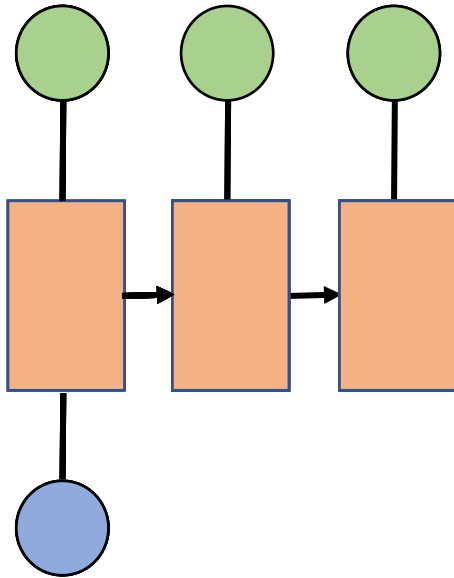
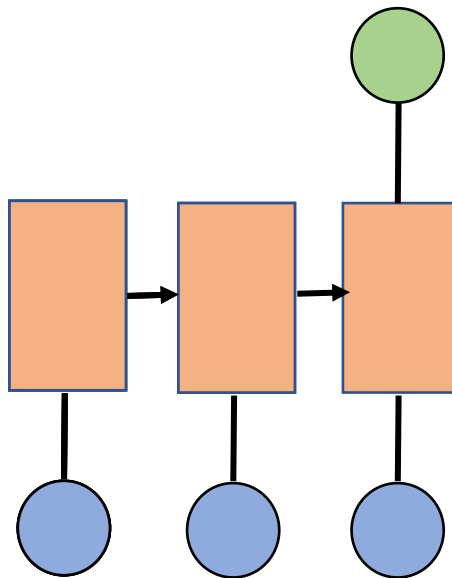


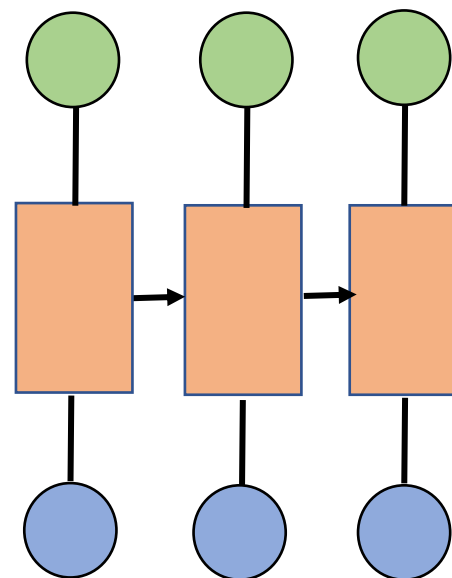
Image
Captioning

many to one



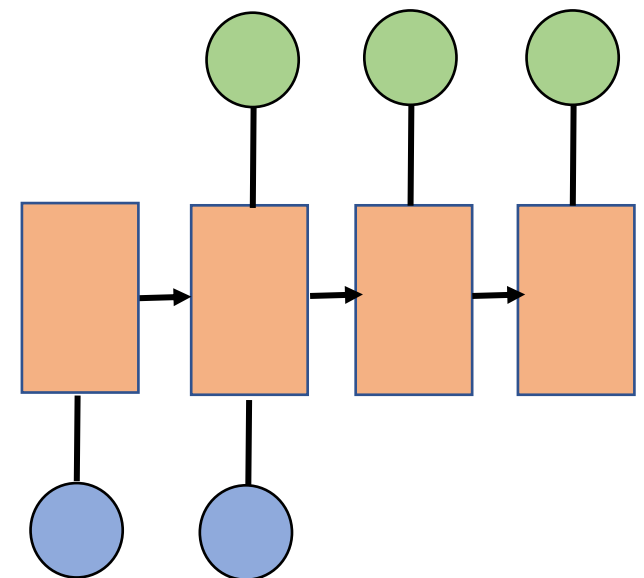
Sentiment
Classification

many to many



Name Entity
recognition

many to many

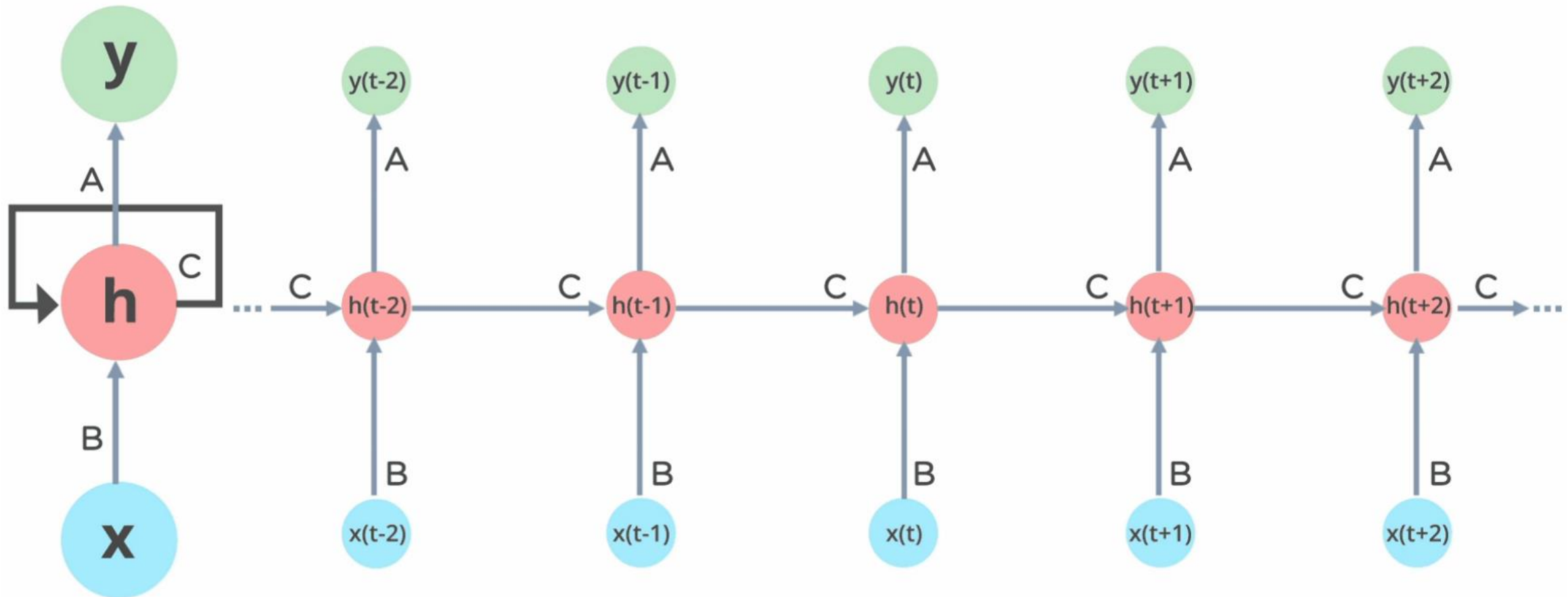


Machine
Translation

Recurrent Neural Network



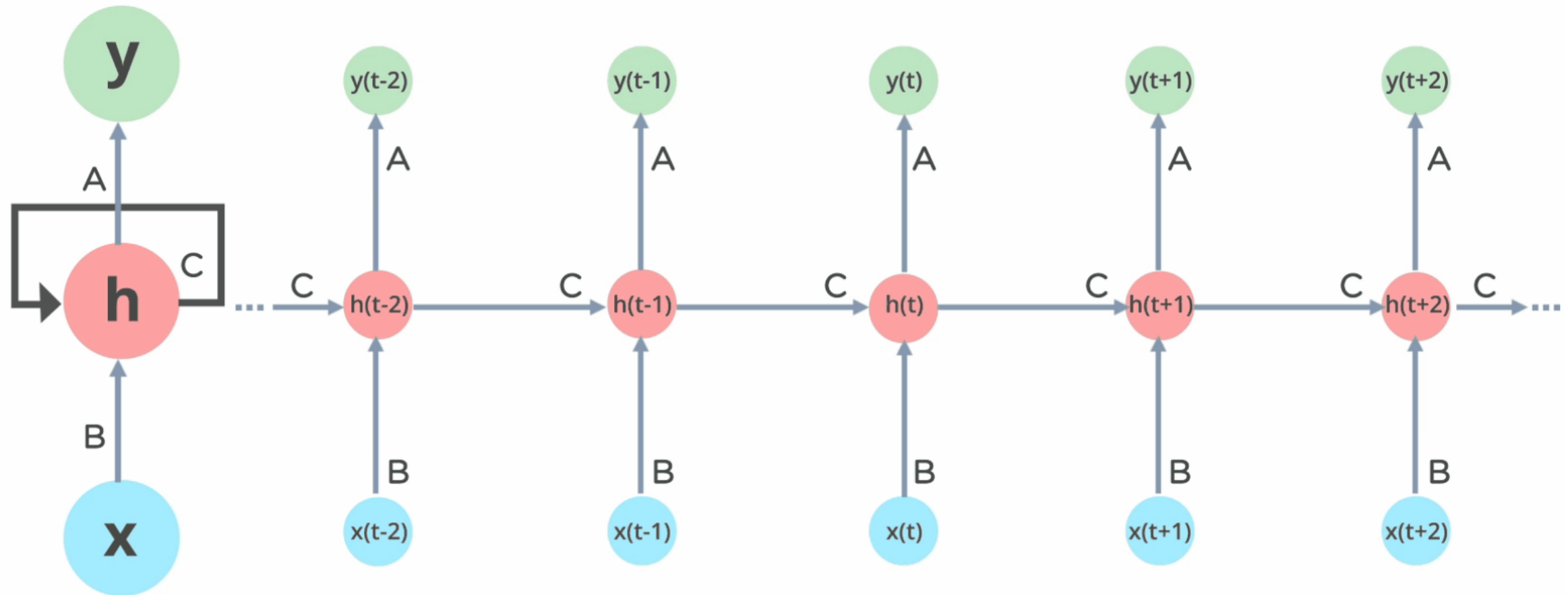
RNN: Forward Propagation



Source: <https://www.simplilearn.com/tutorials/deep-learning-tutorial/rnn>



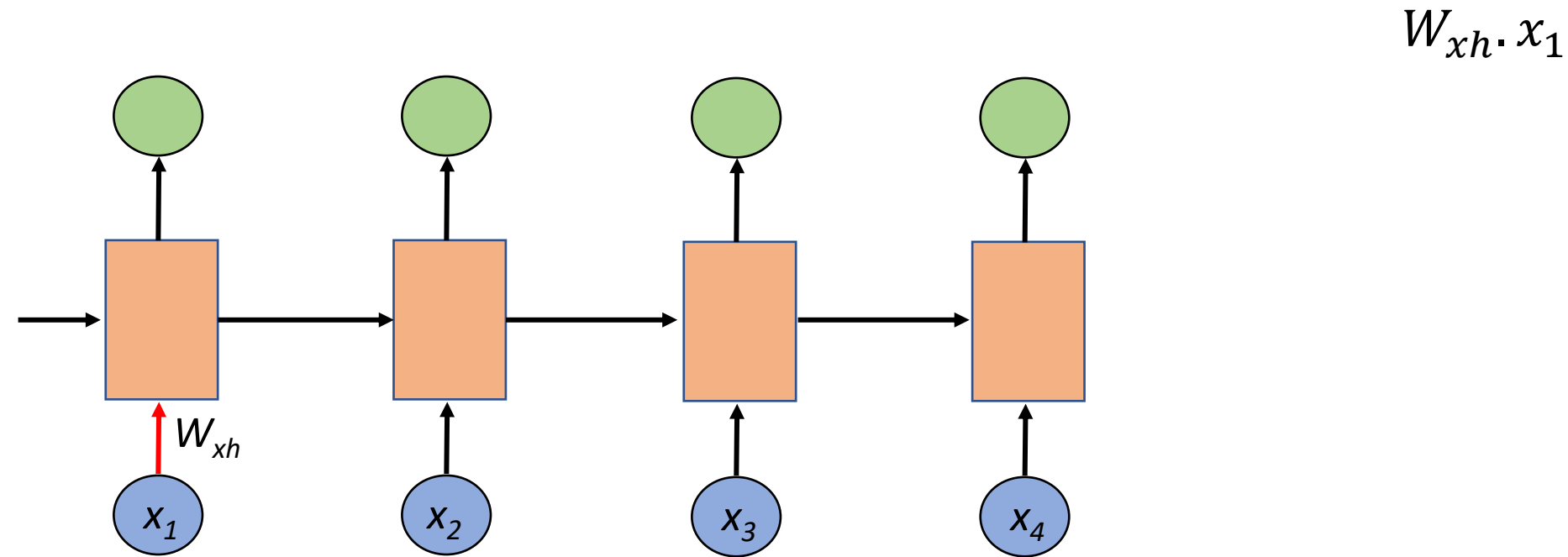
RNN: Forward Propagation



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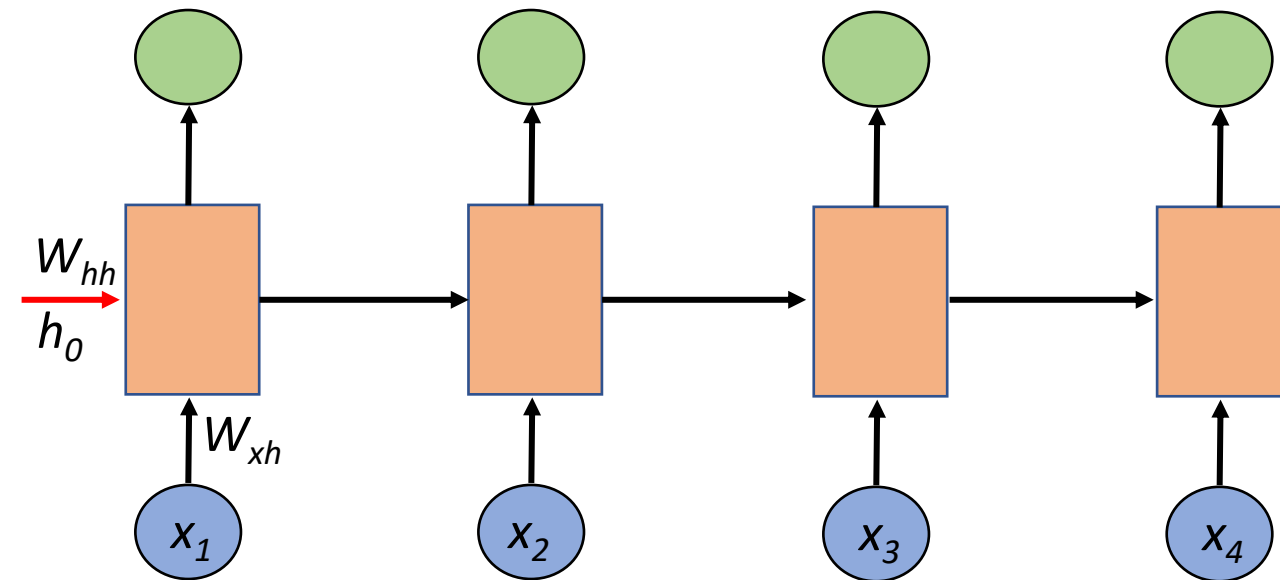
RNN: Forward Propagation





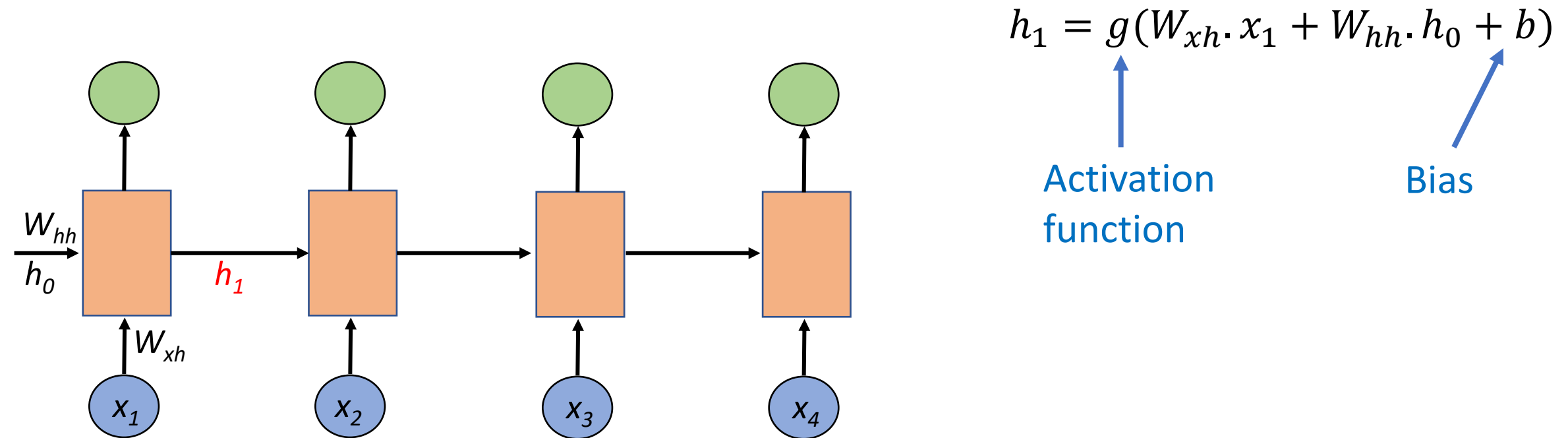
RNN: Forward Propagation

$$W_{xh} \cdot x_1 + W_{hh} \cdot h_0$$



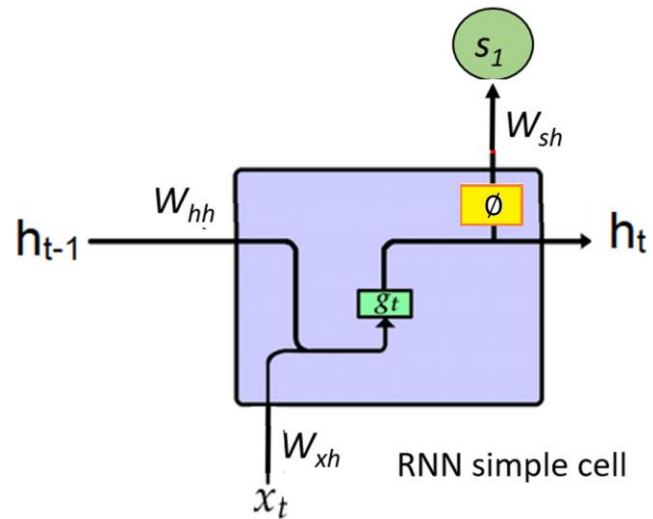


RNN: Forward Propagation



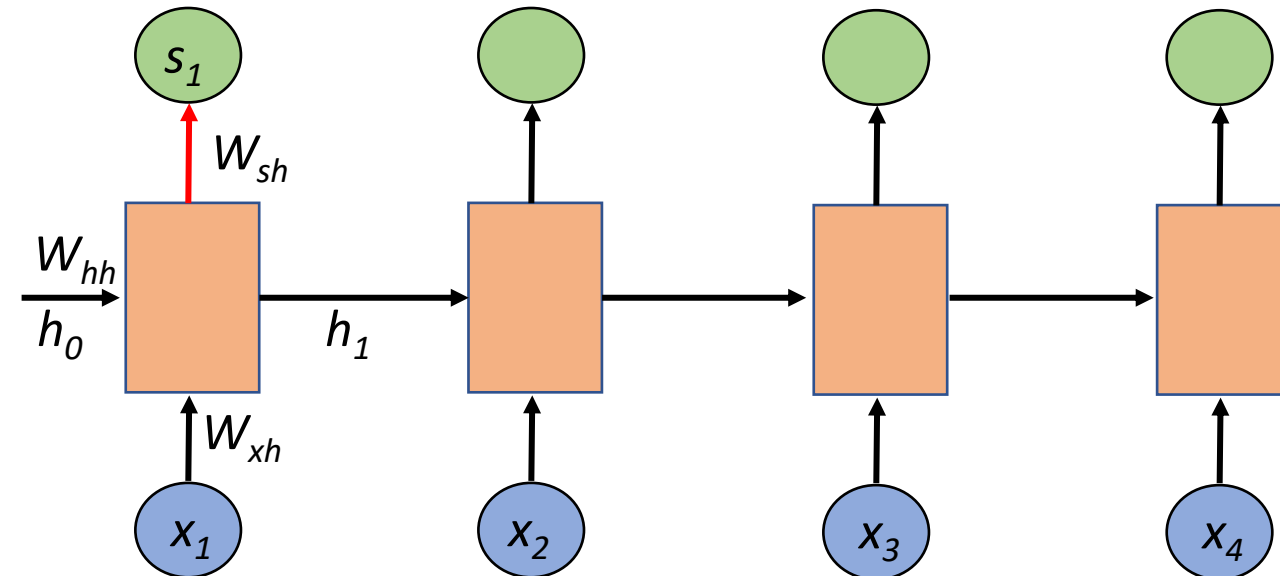


RNN: Forward Propagation



$$h_1 = g(W_{xh} \cdot x_1 + W_{hh} \cdot h_0 + b)$$

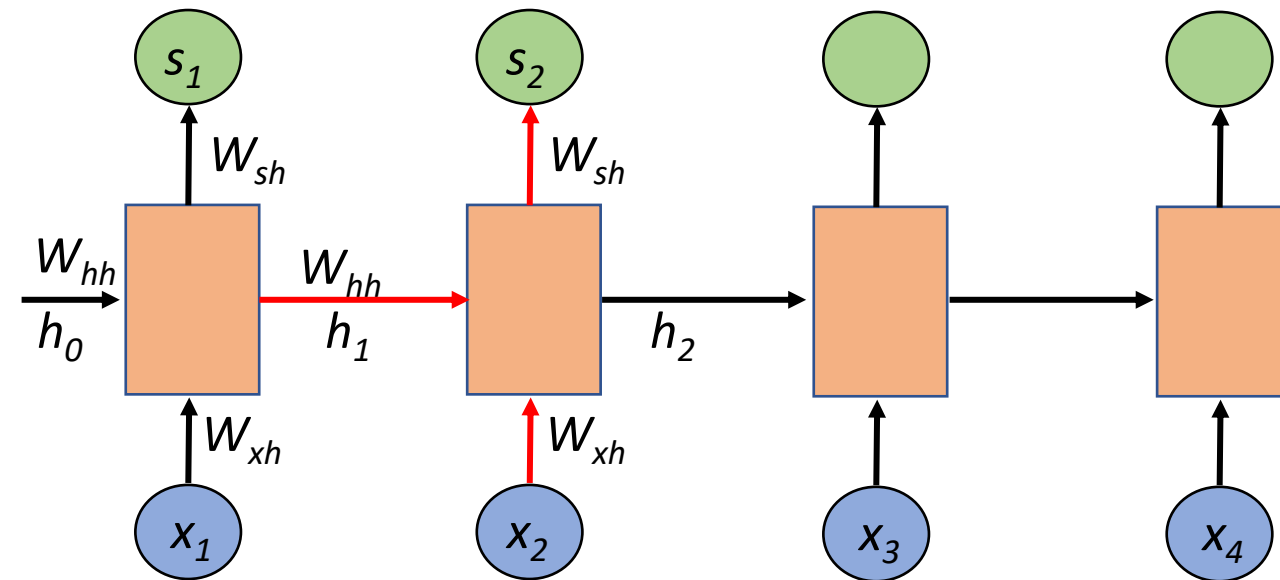
$$s_1 = \phi(W_{sh} \cdot h_1 + c)$$



x_1	Input
h_0	Previous hidden state
h_1	Current hidden state
W_{xh}, W_{hh}, W_{sh}	Shared parameters
g, ϕ	Activation functions
b, c	Biases
s_1	Output state



RNN: Forward Propagation



$$h_1 = g(W_{xh} \cdot x_1 + W_{hh} \cdot h_0 + b)$$

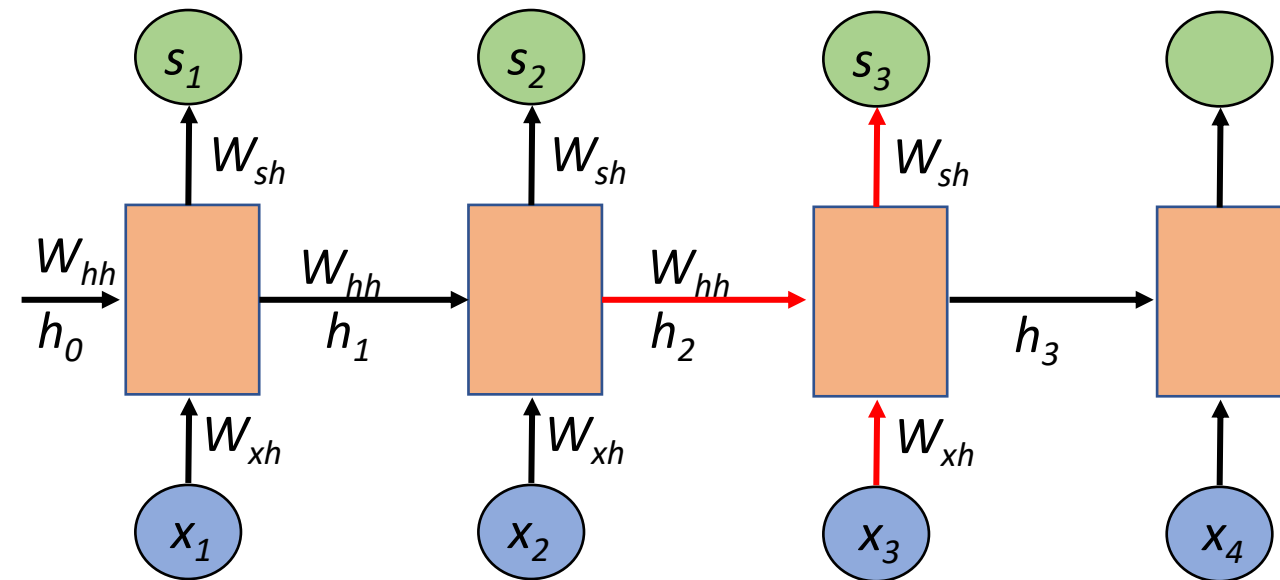
$$s_1 = \phi(W_{sh} \cdot h_1 + c)$$

$$h_2 = g(W_{xh} \cdot x_2 + W_{hh} \cdot h_1 + b)$$

$$s_2 = \phi(W_{sh} \cdot h_2 + c)$$



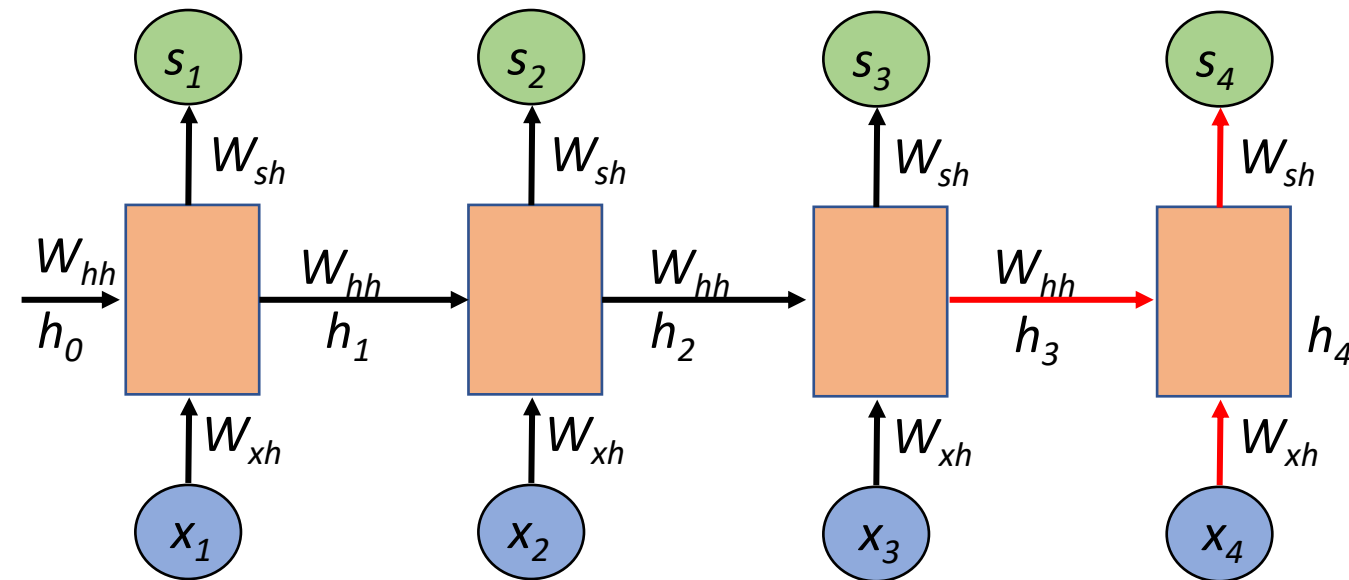
RNN: Forward Propagation



$$\begin{aligned} h_1 &= g(W_{xh} \cdot x_1 + W_{hh} \cdot h_0 + b) \\ s_1 &= \phi(W_{sh} \cdot h_1 + c) \\ h_2 &= g(W_{xh} \cdot x_2 + W_{hh} \cdot h_1 + b) \\ s_2 &= \phi(W_{sh} \cdot h_2 + c) \\ h_3 &= g(W_{xh} \cdot x_3 + W_{hh} \cdot h_2 + b) \\ s_3 &= \phi(W_{sh} \cdot h_3 + c) \end{aligned}$$



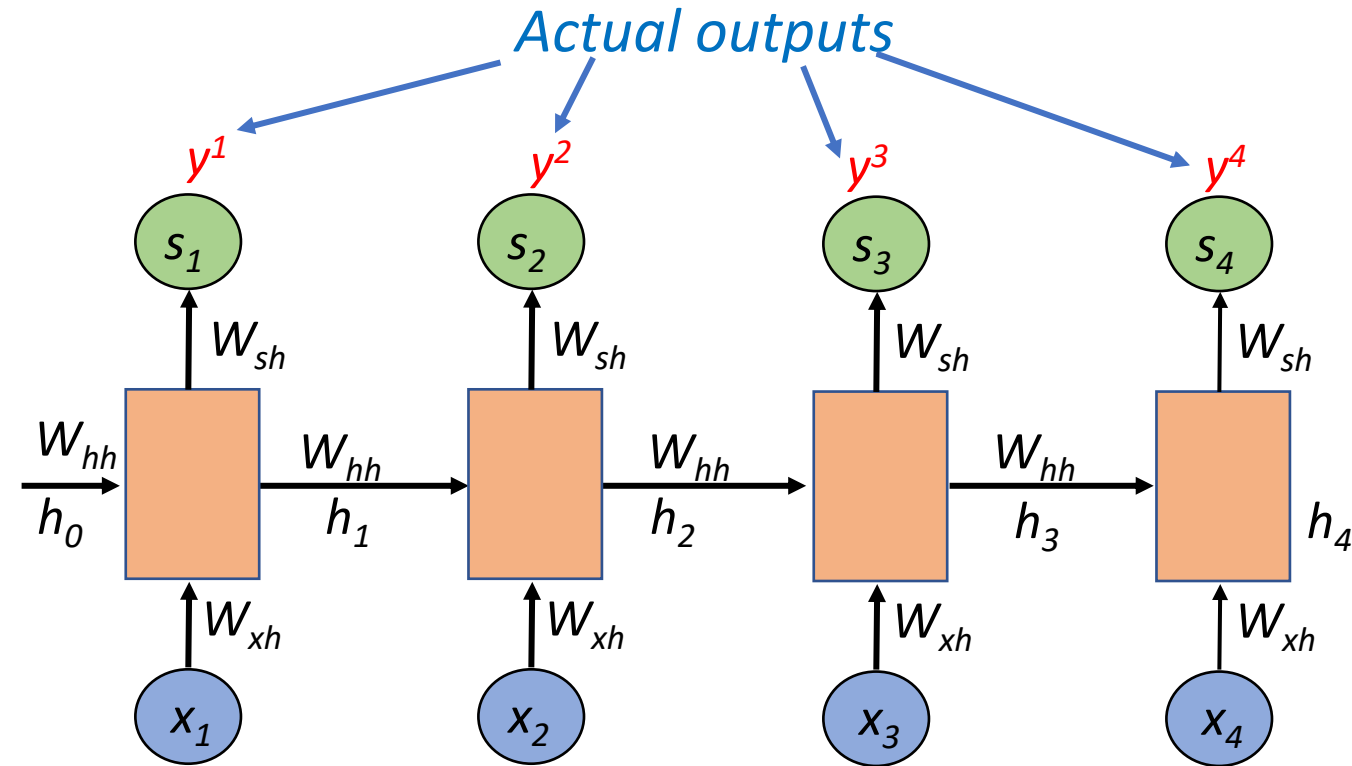
RNN: Forward Propagation



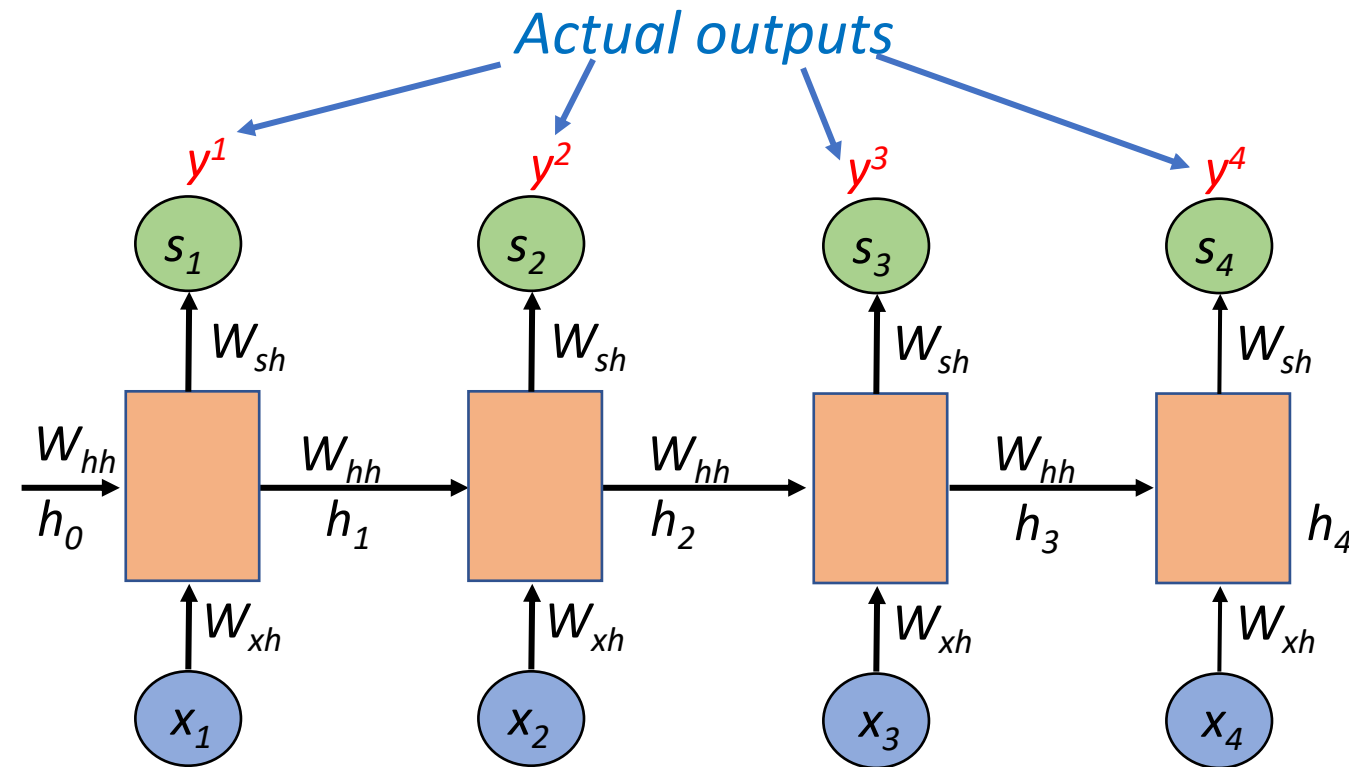
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Weight matrix W_{xh} , W_{hh} and W_{sh} remains same throughout the forward propagation, thus ensuring parameter sharing

RNN: Back Propagation Through Time (BPTT)



RNN: Back Propagation Through Time (BPTT)



Loss calculation:

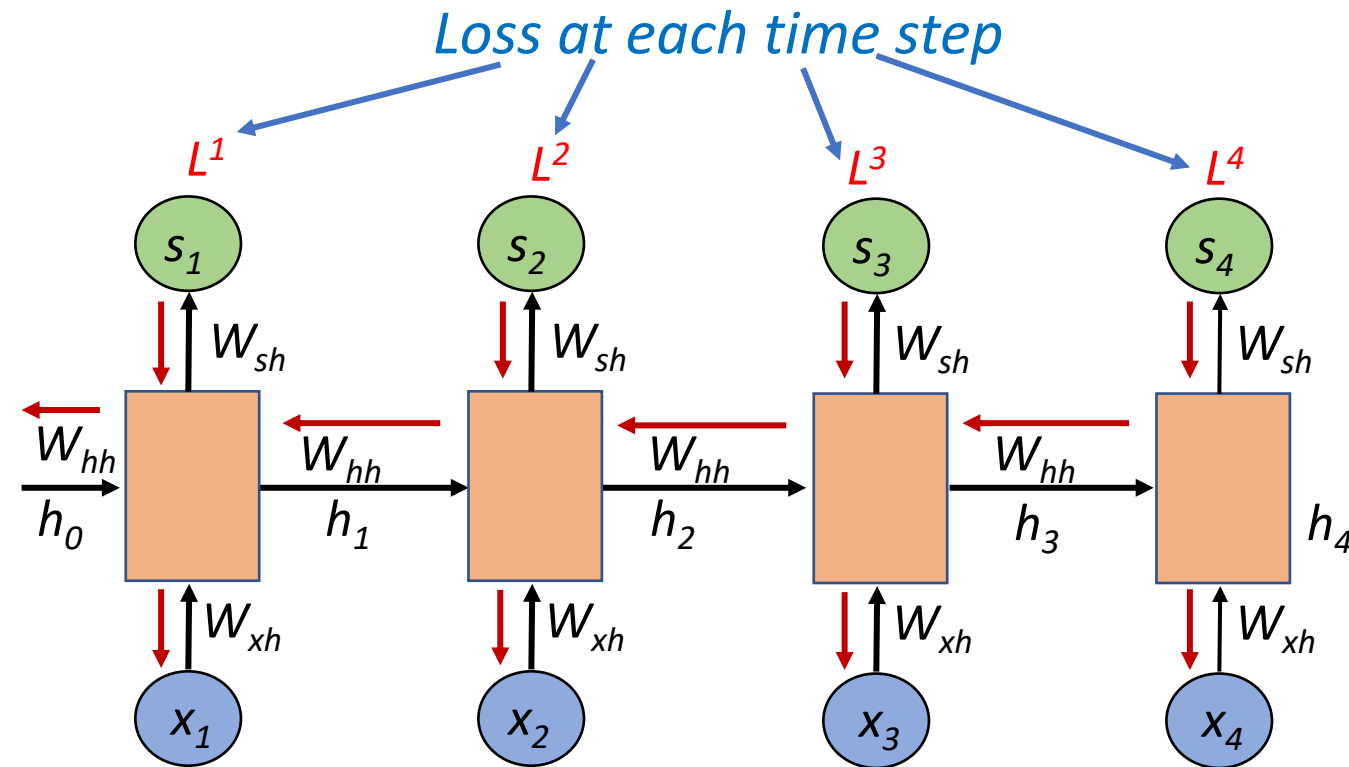
$$\text{Loss } L = \mathcal{L}(y_1, s_1) + \mathcal{L}(y_2, s_2) + \mathcal{L}(y_3, s_3) + \mathcal{L}(y_4, s_4)$$

$\mathcal{L}$ = Loss function

In general:

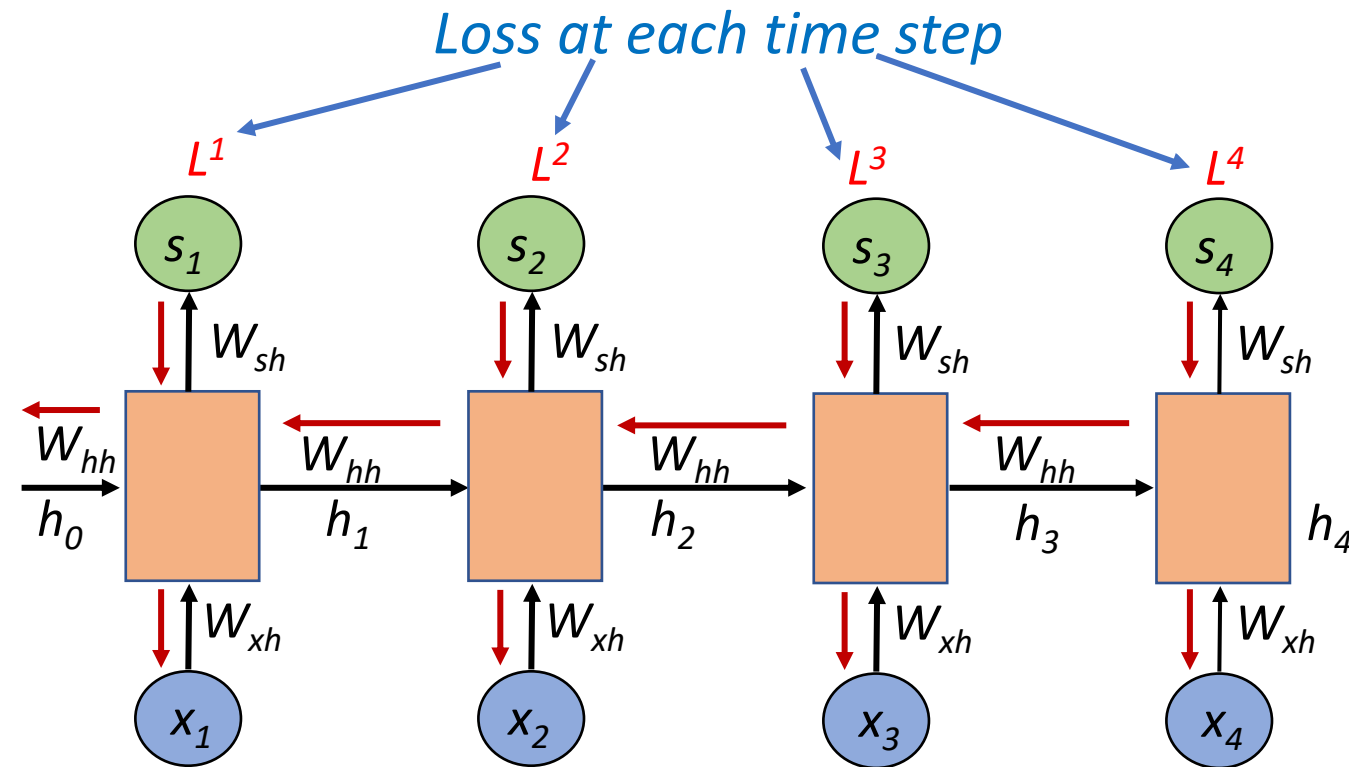
$$L = \sum_{i=1}^n \mathcal{L}(y_i, s_i)$$

RNN: Back Propagation Through Time (BPTT)



Gradient calculation wrt W_{sh} :

RNN: Back Propagation Through Time (BPTT)

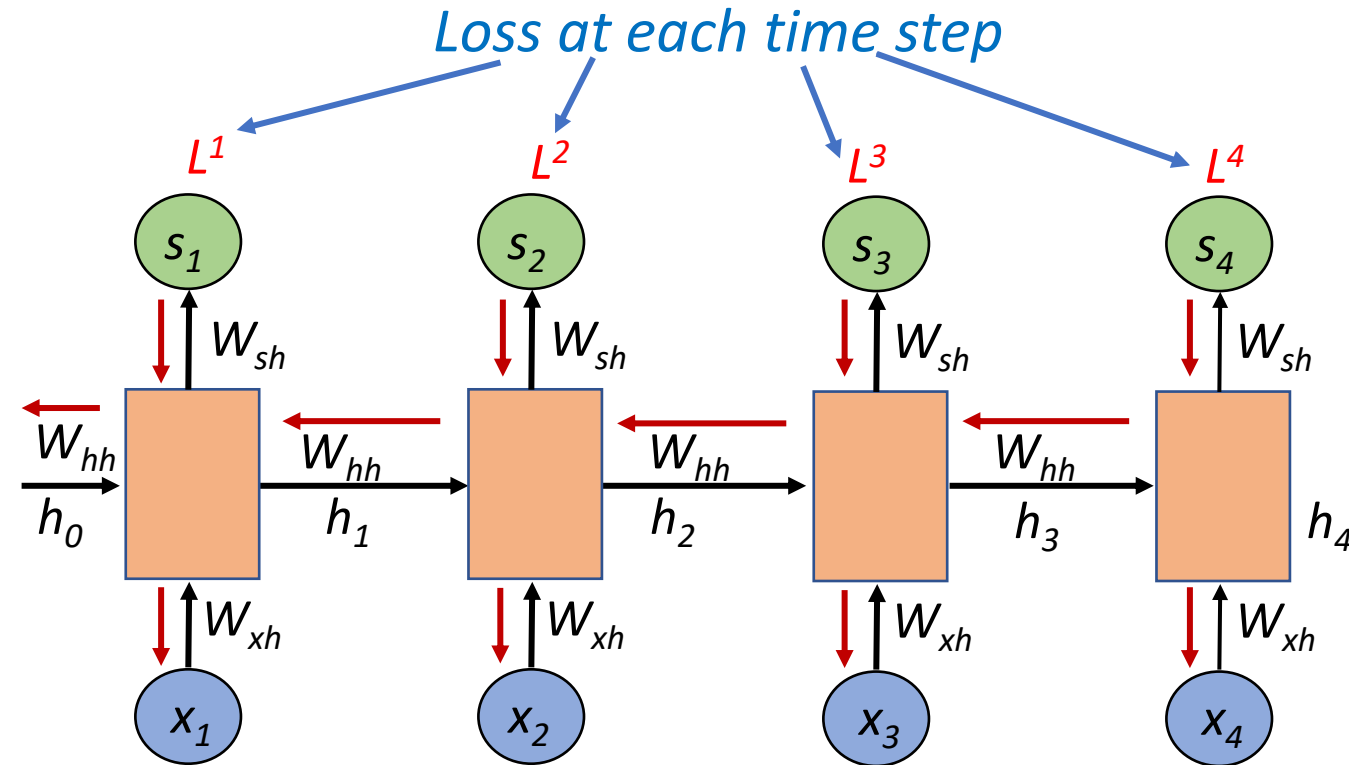


Gradient calculation wrt W_{sh} :

Assumptions:

$$s_4 = W_{sh} \cdot h_4 \text{ and } \mathcal{L} = \text{least square function} = \frac{1}{2} (y_4 - s_4)^2$$

RNN: Back Propagation Through Time (BPTT)



Gradient calculation wrt W_{sh} :

$$\frac{\partial L^4}{\partial W_{sh}} = \frac{\partial L^4}{\partial s_4} \cdot \frac{\partial s_4}{\partial W_{sh}}$$

$$= -(y_4 - s_4) \cdot h_4$$

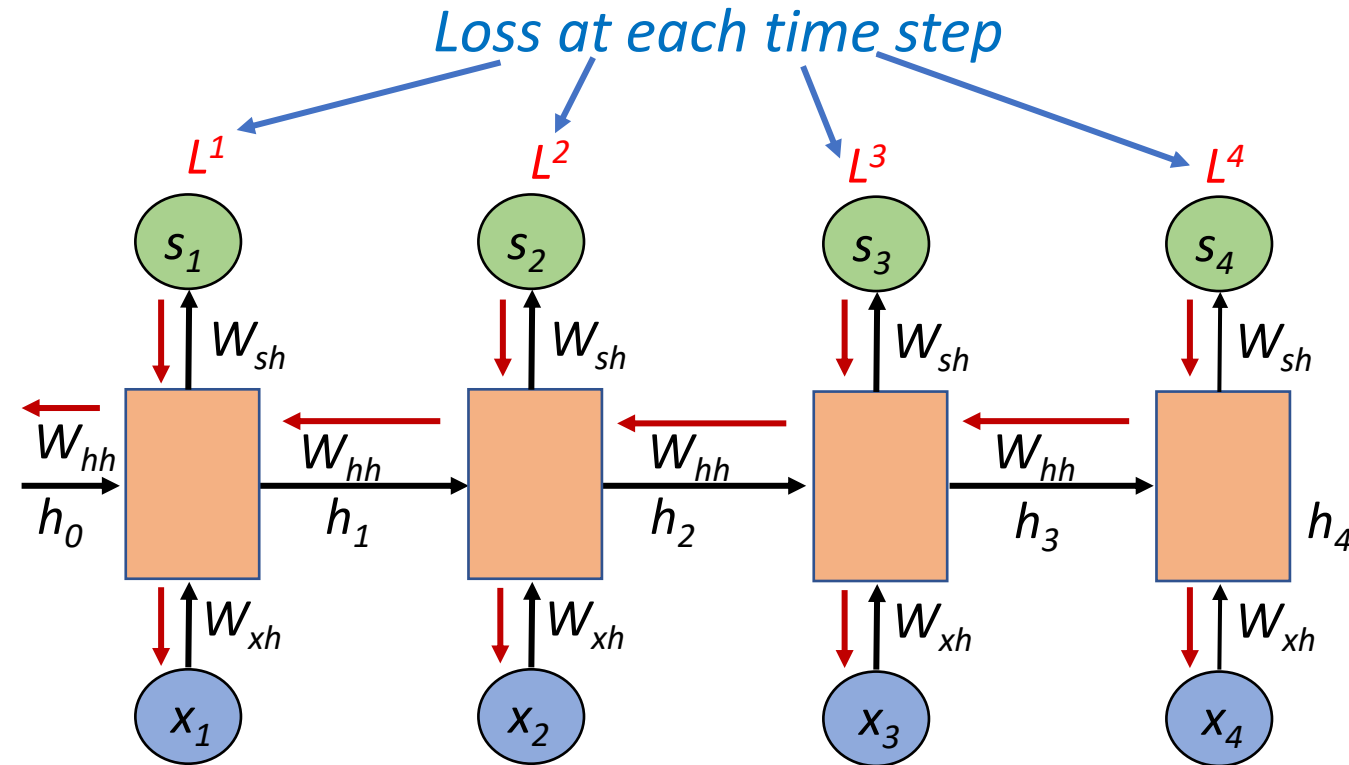
Weight updation wrt W_{sh} :

$$W_{sh} = W_{sh} - \sum_{i=1}^n (y_i - s_i) \cdot h_i$$

Assumptions:

$$s_4 = W_{sh} \cdot h_4 \text{ and } \mathcal{L} = \text{least square function} = \frac{1}{2} (y_4 - s_4)^2$$

RNN: Back Propagation Through Time (BPTT)



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Gradient calculation wrt W_{hh} :

$$\frac{\partial L^4}{\partial W_{hh}} = \frac{\partial L^4}{\partial s_4} \cdot \frac{\partial s_4}{\partial h_4} \cdot \frac{\partial h_4}{\partial W_{hh}}$$

$$= -(y_4 - s_4) \cdot W_{sh} \cdot \frac{\partial h_4}{\partial W_{hh}}$$

Now, $h_4 = g(W_{xh} \cdot x_4 + W_{hh} \cdot h_3)$

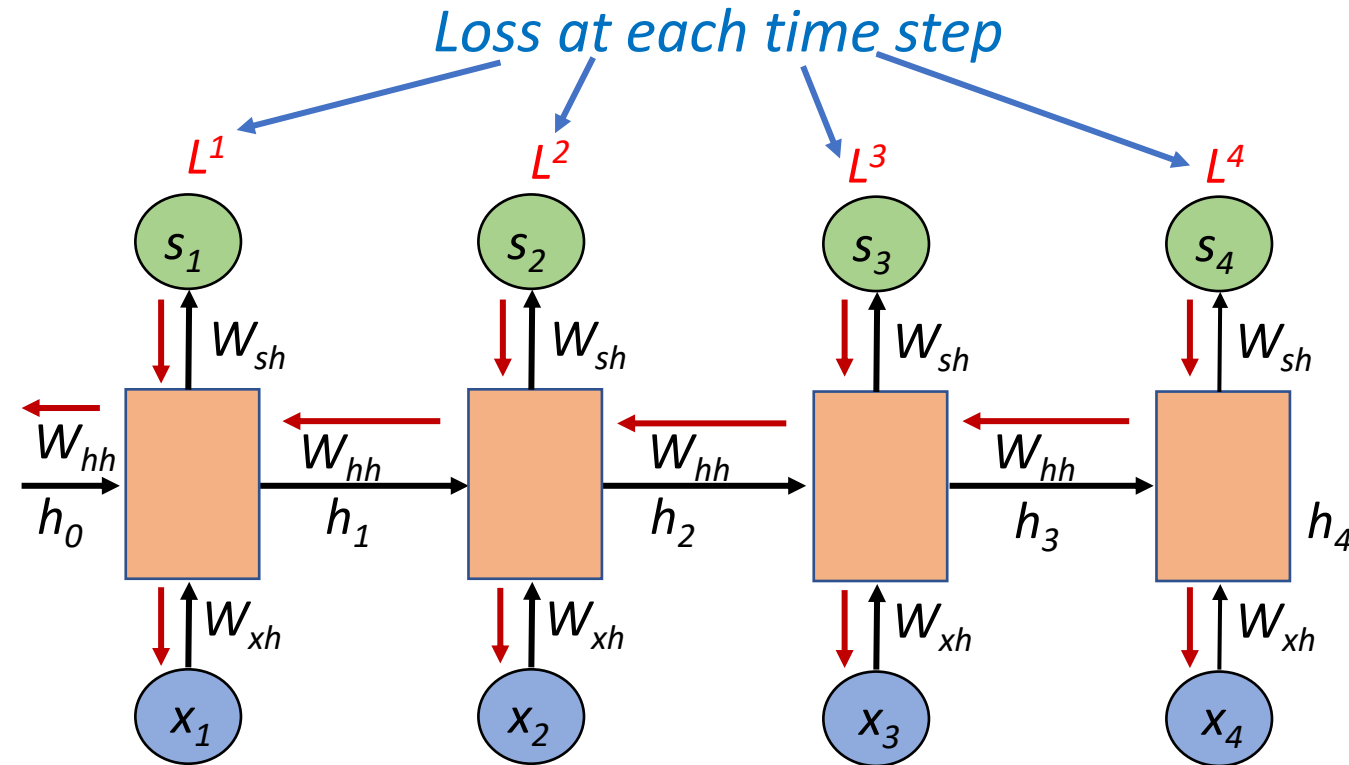
Let, $h_4 = g(z_4)$

$$\text{Then, } \frac{\partial h_4}{\partial W_{hh}} = \frac{\partial h_4}{\partial z_4} \cdot \frac{\partial z_4}{\partial W_{hh}}$$

$$= g'(z_4) \cdot [h_3 + W_{hh} \frac{\partial h_3}{\partial W_{hh}}]$$

$$= g'(z_4) \cdot [h_3 + W_{hh} (h_2 + W_{hh} \frac{\partial h_2}{\partial W_{hh}})]$$

RNN: Back Propagation Through Time (BPTT)



Gradient calculation wrt W_{hh} :

$$\frac{\partial L^4}{\partial W_{hh}} = -(y_4 - s_4) \cdot W_{sh} \cdot g'(z_4) \cdot [h_3 + W_{hh} (h_2 + W_{hh} \frac{\partial h_2}{\partial W_{hh}})]$$

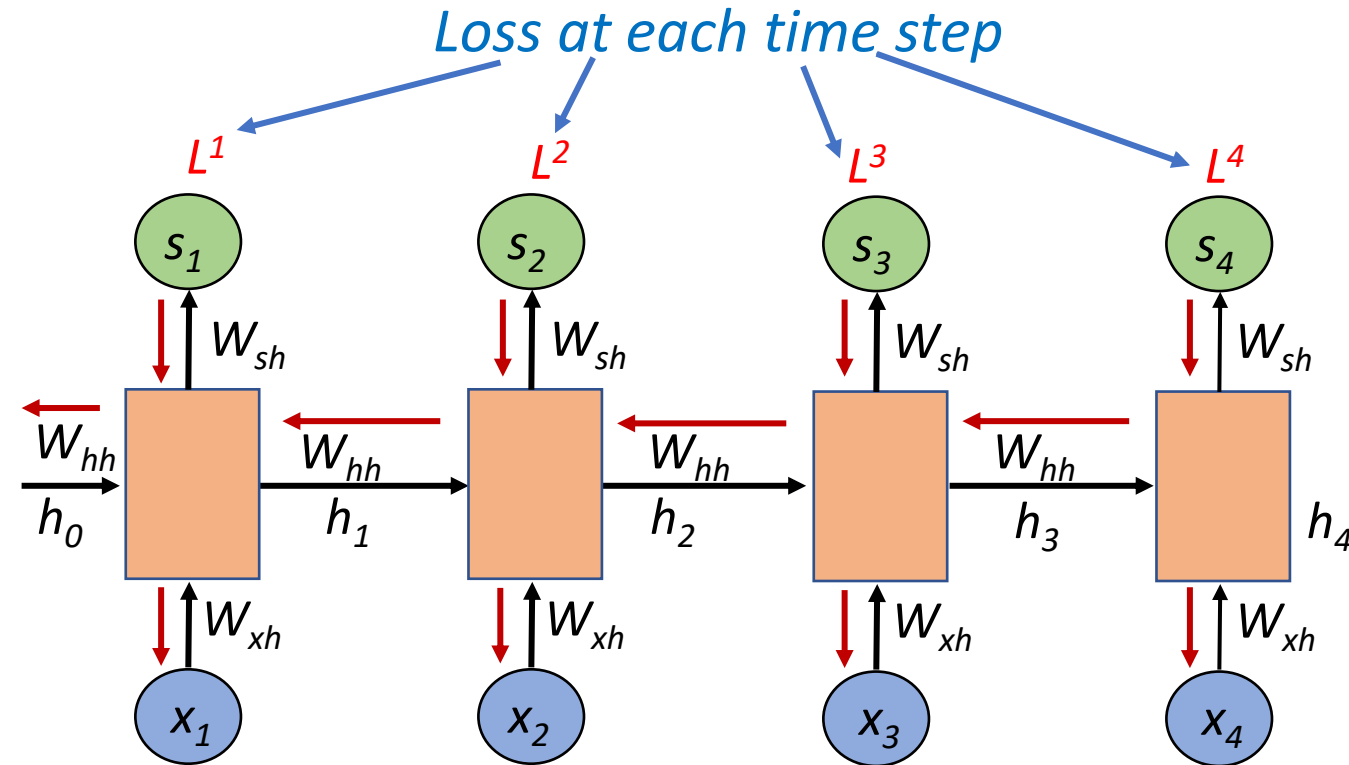
Weight updation wrt W_{hh} :

$$W_{hh} = W_{hh} - \sum_{i=1}^n \frac{\partial L^i}{\partial W_{hh}}$$

Assumptions:

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$$= -(y_4 - s_4) \cdot W_{sh} \cdot \frac{\partial h_4}{\partial W_{xh}}$$

Now, $h_4 = g(W_{xh} \cdot x_4 + W_{hh} \cdot h_3)$

Simply, $h_4 = g(z_4)$

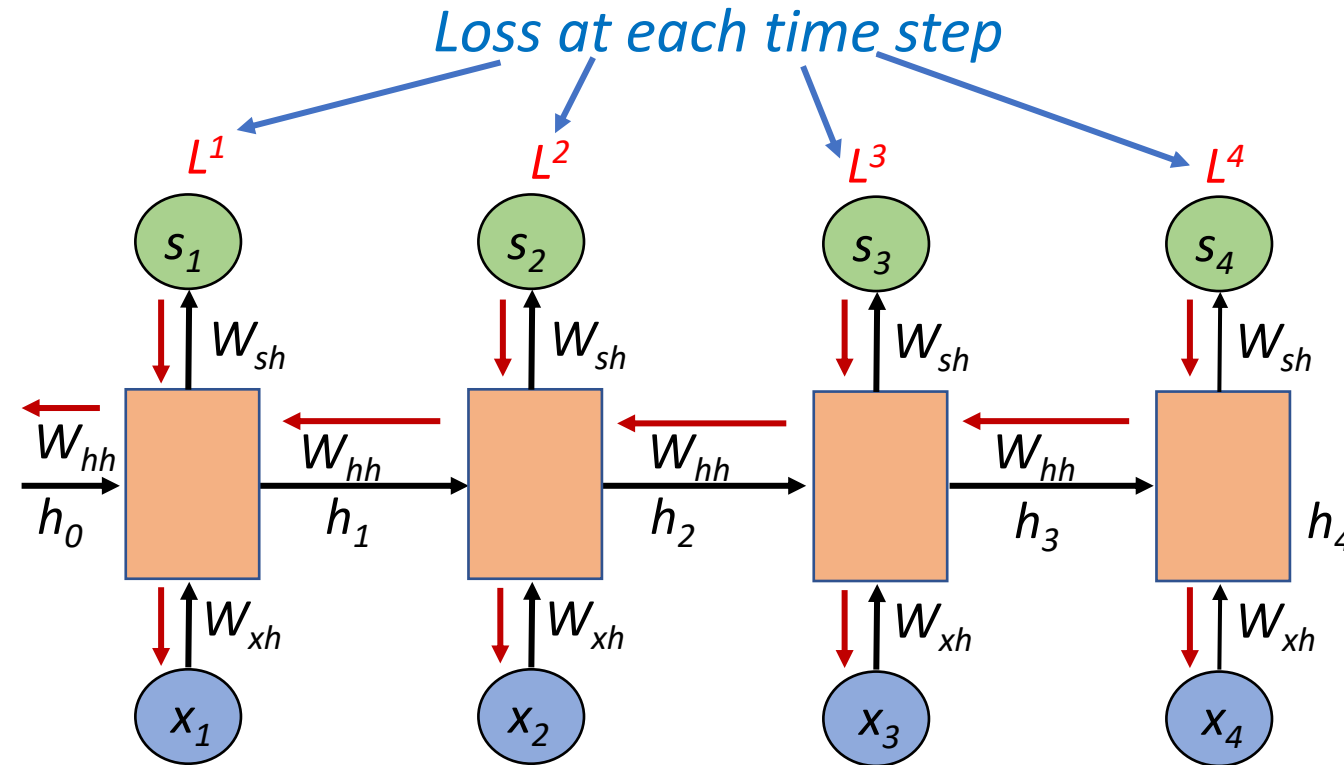
$$\text{Then, } \frac{\partial h_4}{\partial W_{xh}} = \frac{\partial h_4}{\partial z_4} \cdot \frac{\partial z_4}{\partial W_{xh}}$$

$$= g'(z_4) \cdot [x_4 + \frac{\partial W_{hh} \cdot h_3}{\partial W_{xh}}]$$

Assumptions:

$$s_4 = W_{sh} \cdot h_4 \text{ and } \mathcal{L} = \text{least square function} = \frac{1}{2} (y_4 - s_4)^2 = g'(z_4) \cdot [x_4 + W_{hh} \cdot g'(z_3) \cdot \frac{\partial z_3}{\partial W_{xh}}]$$

RNN: Back Propagation Through Time (BPTT)



Gradient calculation wrt W_{xh} :

$$\frac{\partial L^4}{\partial W_{xh}} = -(y_4 - s_4)W_{sh} \cdot g' \cdot [x_4 + W_{hh} \cdot g'(z_2) \cdot \frac{\partial z_2}{\partial W_{xh}} \dots]$$

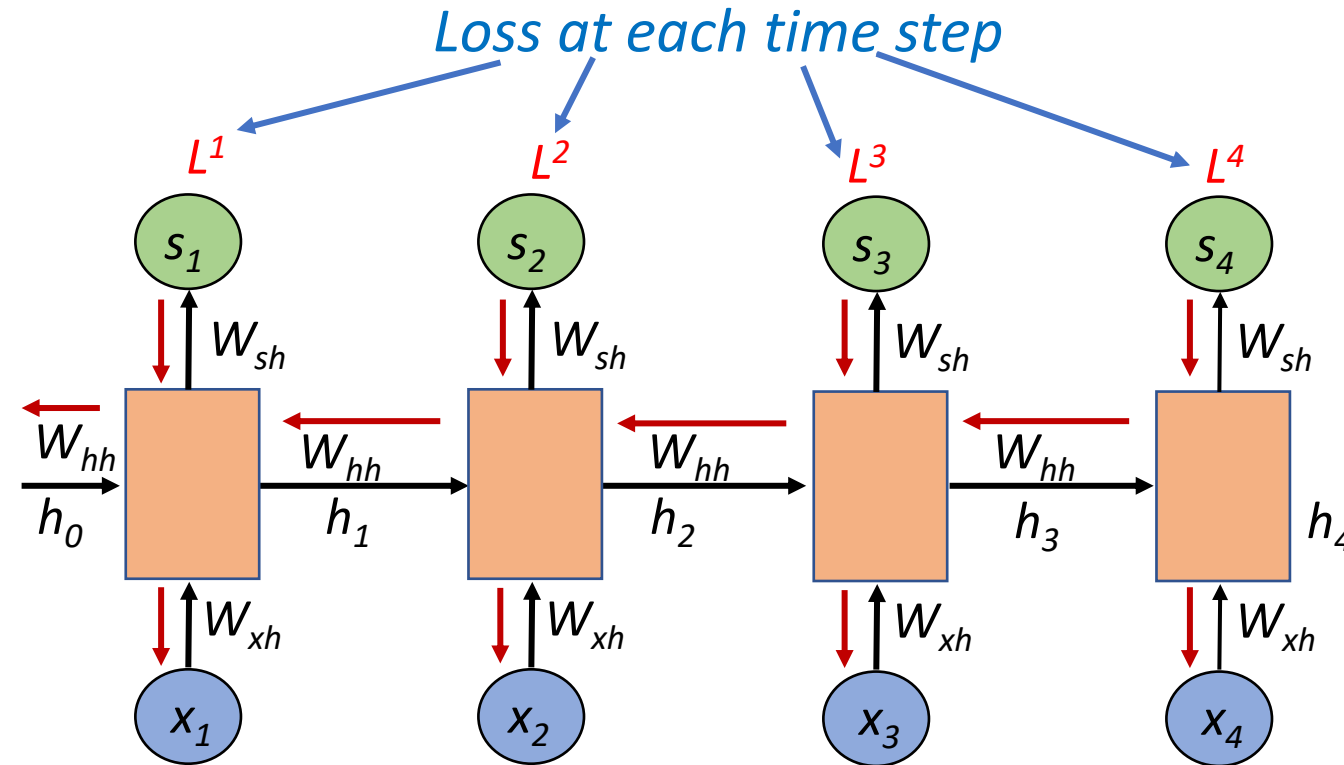
Weight updation wrt W_{xh} :

$$W_{xh} = W_{xh} - \sum_{i=1}^n \frac{\partial L^i}{\partial W_{xh}}$$

Assumptions:

$$s_4 = W_{sh} \cdot h_4 \text{ and } \mathcal{L} = \text{least square function} = \frac{1}{2} (y_4 - s_4)^2$$

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Gradient calculation wrt W_{xh} :

$$\frac{\partial L^4}{\partial W_{xh}} = -(y_4 - s_4) W_{sh} \cdot g' \cdot [x_4 + W_{hh} \cdot g'(z_2) \cdot \frac{\partial z_2}{\partial W_{xh}} \dots]$$

Weight updation wrt W_{xh} :

$$W_{xh} = W_{xh} - \sum_{i=1}^n \frac{\partial L^i}{\partial W_{xh}}$$

`tf.keras.layers.SimpleRNN(rnn_units)`



Assumptions:

$$s_4 = W_{sh} \cdot h_4 \text{ and } \mathcal{L} = \text{least square function} = \frac{1}{2} (y_4 - s_4)^2$$



Limitations of RNN

- Gradient calculation involves many factors of weights and contribution of activation function.





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- This may lead to:

Exploding Gradient



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$$\frac{\partial L}{\partial W} \gg 1$$

→ resulting in NAN values

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Exploding Gradient

$$\frac{\partial L}{\partial W} \gg 1$$

→ resulting in NAN values

Vanishing Gradient

$$\frac{\partial L}{\partial W} \ll 1$$

→ resulting in 0 values



Limitations of RNN

- Gradient calculation involves many factors of weights and contribution of activation function.
- This may lead to:

Exploding Gradient

Vanishing Gradient

- make learning unstable



Limitations of RNN

- Gradient calculation involves many factors of weights and contribution of activation function.
- This may lead to:

Exploding Gradient

- make learning unstable

Vanishing Gradient

- Short term dependencies

“the stars shine in the ?” → sky (*RNN works good here*)

- Long term dependencies

“I grew up in Spain..... I speak fluent Spanish”. (*Difficult for RNN to remember as gap increases*)



Possible Solutions

Exploding Gradient

Vanishing Gradient

❑ Gradient clipping

```
# inside the optimizer we are doing clipping  
optimizer=tf.keras.optimizers.SGD(clipvalue=0.5)
```



Possible Solutions

Exploding Gradient

- ☐ Gradient clipping

```
# inside the optimizer we are doing clipping  
optimizer=tf.keras.optimizers.SGD(clipvalue=0.5)
```

Vanishing Gradient

- ☐ Activation function (Relu)
- ☐ Weight initialization (identity matrix)
- ☐ Gated cells (LSTM,GRU,etc)